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A hybrid deep learning–ensemble model for accurate and explainable electricity demand forecasting in renewable energy communities

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ABSTRACT

Accurate, interpretable, and scalable short-term electricity demand forecasting is a crucial requirement for transitioning towards decentralized, citizen-driven energy systems, particularly within renewable energy communities (RECs). The proposed framework utilizes long short-term memory (LSTM) networks and extreme gradient boosting (XGBoost) to capture both the sequential dynamics of consumption and the nonlinear interactions between household and community-level characteristics. A stratified household selection strategy based on contribution-level binning is employed to ensure coverage across distinct demand regimes while maintaining computational efficiency. Based on 15-min smart meter data from a renewable energy community in Loureiro, Portugal, the hybrid model achieved the lowest prediction errors at both scales, with a best-case household-level mean absolute percentage error (MAPE) of 1.25% and a community-level average MAPE of 3.14%, corresponding to explained variance (R^2) values of 0.9420 and 0.9558, respectively. Although XGBoost performed slightly better for specific low-variability households, the hybrid approach consistently outperformed the standalone models in terms of accuracy and robustness. The model's interpretation is enhanced by SHAP, which reveals distinct demand drivers for consumers, including memory-based characteristics and climatic variables like temperature and solar radiation. The Interpretive, multi-level prediction framework provides a reliable basis for practical REC applications, such as battery planning, peer-to-peer trading, and real-time demand response.

1. Introduction

Climate change, the defining global challenge of the 21st century, has prompted international frameworks such as the Paris Agreement and the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report to call for the rapid decarbonization of energy systems. At the regional level, the European Union (EU) has made a significant commitment to this cause with the launch of the European Green Deal in 2019. This ambitious initiative aims to decouple economic growth from resource consumption and make the EU the first climate-neutral continent by 2050 [1].

The building sector, a key player in the transition to renewable energy, is responsible for more than 40% of annual energy consumption and 36% of total carbon dioxide (CO_2) emissions globally [2]. These figures underscore the sector's pivotal role in implementing advanced demand-side strategies, including digital control, decentralization, and short-term forecasting, to enhance efficiency and flexibility. The EU's increasing promotion of policy frameworks that encourage local energy

List of Abbreviations: ANN, Artificial Neural Network; ARIMA, AutoRegressive Integrated Moving Average; Bi-LSTM, Bi-directional Long Short-Term Memory; CEMS, Community Energy Management System; CNN, Convolutional Neural Network; DL, Deep Learning; EMD, Empirical Mode Decomposition; ELF, Electric Load Forecasting; EU, European Union; GRU, Gated Recurrent Unit; HVAC, Heating, Ventilation, and Air Conditioning; kWh, Kilowatt-hour; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; MAPE, Mean Absolute Percentage Error; ML, Machine Learning; MLP, Multi-Layer Perceptron; MSE, Mean Squared Error; PV, Photovoltaic; REC, Renewable Energy Community; RF, Random Forest; RNN, Recurrent Neural Network; RMSE, Root Mean Squared Error; SHAP, SHapley Additive exPlanations; SVM, Support Vector Machine; SVR, Support Vector Regression; IID, Independent and Identically Distributed; P2P, Peer-to-Peer; XAI, Explainable Artificial Intelligence.

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participation and the development of distributed energy resources further underscores the sector's importance. In this context, one of the key policy developments was the legal recognition of Renewable Energy Communities (RECs) under the Clean Energy for All Europeans package and the recast Renewable Energy Directive (RED II, 2018) [3,4]. RECs are formally defined entities composed of citizens, municipalities, local authorities, and small and medium-sized enterprises that participate voluntarily in the generation and management of renewable energy [5]. Unlike traditional utility companies, RECs prioritize social, economic, and environmental value over profit [6]. Through cooperative ownership models and local governance, RECs enhance citizen engagement, promote local energy autonomy, and enable more sustainable community development.

For these communities to function effectively, they require digital and forecasting capabilities within control systems that support the integration and optimisation of distributed energy resources. Their reliance on decentralised assets, such as heating units, photovoltaics, and battery storage, introduces significant operational complexity and unpredictability. The digital infrastructure for managing these resources is provided by Community Energy Management Systems (CEMS), which enable real-time monitoring, coordination, and control at both household and community levels [7]. Within these systems, short-term load forecasting (STLF) plays a foundational role by supporting proactive energy allocation, load shifting, storage scheduling, and local balancing. High-resolution smart meter data provide the basis for modeling dynamic and nonlinear electricity consumption patterns using statistical, machine learning, and deep learning approaches. Forecasting not only enhances operational efficiency but also strengthens system stability, reduces transmission losses, and improves overall flexibility [8,9].

Recent advances in short-term forecasting have explored hybrid architectures that integrate recurrent and ensemble methods to capture temporal patterns and complex feature interactions. While such models have demonstrated improved performance in smart grid settings, their direct application to RECs remains limited, particularly where computational feasibility, user-level variability, and real-time control constraints are critical.

In parallel, increasing attention has been placed on the interpretability of data-driven methods, reflecting the operational and social transparency requirements of community-based energy systems. Addressing these concerns, this study proposes a deployment-oriented forecasting framework for RECs. The contribution of this study lies in the deployment-oriented integration of contribution-based representative household selection, multi-level forecasting, temporal robustness evaluation, and interpretability within a unified framework tailored to REC applications. The framework operates at both household and community aggregation levels to capture individual consumption variability while leveraging aggregation effects relevant to local balancing and Community Energy Management Systems (CEMS). A key methodological contribution of the study is a contribution-based representative household selection protocol that prioritises operational transparency and deployment feasibility. By assigning households to explicit contribution bins, the approach preserves distinct low-, medium-, and high-demand regimes and provides a directly interpretable, traceable selection rationale, which is advantageous for reproducibility and stakeholder interpretation in REC settings. Within this workflow, sequential LSTM-XGBoost modelling and month-wise validation are adopted as forecasting and evaluation components to support heterogeneous and temporally variable demand conditions. In addition, SHAP-based interpretability facilitates the identification of important demand drivers and supports practical REC operation, including user-level feedback, demand response actions, and battery scheduling. Despite growing interest in short-term energy forecasting, key challenges remain in deploying accurate, interpretable, and scalable models for RECs. These include underrepresentation of user heterogeneity, limited deployment-aligned evaluation, and insufficient operational interpretability in forecasting workflows. To address these gaps, this study is guided by the following research questions:

(i) How can short-term electricity demand be forecasted in RECs at both household and community levels while remaining computationally feasible for real-world deployment?
(ii) To what extent can a contribution-based representative household selection strategy preserve visible demand heterogeneity and individual-level information in a transparent and deployment-feasible manner for REC applications?
(iii) How robust is the proposed forecasting framework under temporal variability, including month-wise and seasonal changes, that characterize real operational conditions in RECs?
(iv) How can model interpretability be integrated to support decision-making and user-level insights in REC forecasting?

To address these questions, we propose a multi-level forecasting framework for RECs that integrates: (1) a contribution-based representative household selection protocol to support transparent coverage of demand regimes; (2) forecasting at both household and community aggregation levels; (3) a month-wise incremental validation protocol to reflect real-world deployment conditions; and (4) SHAP-based interpretability to support operational decision-making. This design ensures that the framework is accurate, interpretable, computationally feasible, and suitable for deployment within decentralised energy communities.

2. Related work

Building upon the motivation for short-term forecasting and decentralized control in RECs, this section reviews recent developments in load prediction models, with an emphasis on hybrid architectures and deployment-oriented strategies. The rapid adoption of smart meters in RECs has created new opportunities for high-resolution electricity load forecasting. In these decentralized energy systems, accurate short-term forecasting is essential for enabling peer-to-peer (P2P) trading, battery scheduling, and demand shaping [8]. However, the complexity of user-specific consumption patterns and their nonlinear dynamics presents challenges for conventional statistical models such as autoregressive integrated moving average or exponential smoothing.

Forecasting electricity demand in RECs is primarily a time-series problem, typically addressed using either physical (white-box) or data-driven (black-box) models. Physical models, such as *EnergyPlus* [10], offer interpretability but are limited by their dependence on detailed building inputs and poor scalability across heterogeneous households. In contrast, data-driven models leverage smart meter data to learn complex consumption behaviours, making them more suitable for decentralised settings [11,12]. Machine learning methods such as XGBoost, random forests, and support vector regression are effective in capturing nonlinear relationships in electricity consumption data with relatively low computational overhead and offer interpretability through tools such as Shapley Additive exPlanations (SHAP) [13,14].

Beyond predictive accuracy, interpretability remains a critical requirement in RECs, where energy decisions are often shared among multiple stakeholders. While ensemble learning models such as XGBoost provide inherent transparency through feature attribution techniques like SHAP, most deep learning frameworks operate as black-box models, limiting their suitability for operational decision-making and user trust [14]. Hybrid approaches offer a practical compromise by retaining the explanatory advantages of tree-based models while leveraging deep neural networks to capture complex temporal dynamics, thereby aligning performance with transparency requirements in REC environments.

These complementary strengths and weaknesses have motivated the development of hybrid architectures that integrate recurrent neural networks with ensemble learning methods to achieve robust and scalable short-term load forecasting. Recent studies have introduced hybrid deep learning frameworks for electricity demand forecasting in smart grid environments, demonstrating strong predictive performance using combinations of recurrent and convolutional neural networks [15,16]. While

these models show improvements in short-term accuracy and multi-output capabilities, they are typically designed for centralized systems and do not address the operational constraints or user-level variability found in RECs. Recent advances have introduced Transformer-based forecasting architectures, diffusion probabilistic models, and federated learning frameworks for decentralized energy systems [17,18]. While promising, these methods often involve high computational cost and complexity, and remain underexplored in real-world REC deployments. In contrast, hybrid models offer a more practical balance of accuracy, interpretability, and scalability, particularly in scenarios with limited resources or strict privacy constraints.

Despite recent advances, several critical challenges remain in electricity demand forecasting for RECs. User heterogeneity is frequently underrepresented, as many studies rely on aggregated or unstratified smart meter data, which limits the ability to analyse model behaviour across distinct demand regimes [19]. As a result, forecasting approaches often fail to explicitly assess performance across low-, medium-, and high-contribution users, an aspect that is particularly relevant in decentralised RECs. Although clustering-based approaches have been proposed to address demand heterogeneity [20–22], such methods often introduce additional computational complexity and sensitivity to algorithmic assumptions, reducing transparency and generalisability in deployment-oriented settings. In addition, interpretability continues to pose a significant challenge: while ensemble learning models such as XGBoost support feature attribution, most deep learning frameworks operate as black boxes, constraining their suitability for shared decision-making in REC environments [14]. Further limitations include inconsistent temporal resolutions across studies, limited evaluation under real operational conditions, and a predominant focus on academic or public buildings rather than mixed-use or community-scale contexts [19,23].

However, such approaches typically overlook interpretability and do not incorporate hybrid learning architectures suited for RECs. Moreover, there is limited exploration of probabilistic methods in REC contexts, despite their importance for quantifying uncertainty in volatile consumption environments [24,25]. Finally, hybrid approaches, while promising, have rarely been tested in complex community-scale settings, and most existing studies overlook the integration of energy sharing and optimisation mechanisms essential for CEMS [26]. Lastly, privacy and decentralisation constraints further complicate data access and sharing. While federated learning has emerged as a potential solution, it still faces open challenges related to non-IID data distributions and efficient aggregation mechanisms [26]. Compared to existing hybrid forecasting models, which often focus on grid-level or building-specific datasets and do not account for user-level diversity or interpretability [15,19], the proposed framework integrates stratified household selection, SHAP-based analysis, and month-wise validation to enhance transparency, representativeness, and practical relevance for REC deployment.

While there have been significant advances in the precision and development of forecasting models, important limitations remain in existing approaches, particularly in their ability to capture user-level heterogeneity, a key feature for equitable energy management and planning in RECs. While clustering-based methods have been proposed to segment users, they may reduce operational traceability and can introduce additional tuning complexity, particularly in decentralised environments. Additionally, although interpretability has been explored through ensemble methods such as XGBoost combined with SHAP, deep learning models often remain difficult to interpret, which can reduce their suitability for shared decision-making and operational feedback within RECs. Temporal generalisation is another concern, as most studies continue to rely on static validation protocols, with limited attention to evolving demand patterns or seasonal variation. Finally, while recent work has begun to explore Transformer architectures, federated learning, and probabilistic forecasting, these methods often carry significant computational overhead and remain under-tested in REC-specific deployments.In this context, the present study contributes a deployment-oriented REC forecasting framework that combines contribution-based representative household selection, multi-level forecasting, temporal robustness assessment, and interpretability within a unified operational setting.

As summarized in Table 1, prior studies face limitations related to household-level representativeness, interpretability, computational scalability, and deployment realism. The proposed framework is designed to address these gaps by integrating a contribution-based representative household selection protocol, sequential LSTM–XGBoost modelling, SHAP-based interpretability, and month-wise incremental validation aligned with real-world REC operation. As shown in Table 1, prior studies demonstrate strong performance of data-driven and hybrid forecasting models when high-resolution data are available. However, the majority of existing research remains focused on centralized or building-level applications, with limited attention to deployment feasibility, household-level representativeness, and operational interpretability within RECs. These gaps underscore the need for forecasting frameworks that explicitly address the social, technical, and computational constraints unique to REC environments.

3. Methodology

This paper proposes a multi-level STLF framework for RECs that is intended to operate at the household and community levels. This study's key question is: *How can multi-level modelling that captures both aggregate community-level demand and household-level variability while maintaining computational efficiency and interpretability increase STLF in RECs*? The proposed approach ensures both accuracy and scalability in order to answer this question. The methodology presents a stratified binning strategy that methodically chooses households over the whole demand range, in contrast to clustering-based approaches, which are computationally demanding and difficult to understand. This design ensures that the model's internal mechanisms are transparent and easy to comprehend while simultaneously maintaining behavioural variability. This transparency is critical for maintaining confidence in the model's predictions for REC-scale predictions. The overall pipeline, illustrated in Fig. 1, consists of four sequential steps: data preprocessing, household selection, model development, and evaluation. This study focuses on analyzing the electricity consumption patterns of 172 residential buildings located in Loureiro, Portugal [32]. The closest weather station in Aveiro, Portugal, which operates under comparable weather circumstances, provided climate data, including mean temperature, relative humidity, global solar radiation, and wind speed. The data includes 15-min smart meter electricity consumption for each household and weather related variable values. In order to evaluate temporal robustness, the month-wise incremental validation employs an expanding-window evaluation strategy in which models are trained solely on historical data and evaluated on the month that follows. This process is completely distinct from the fixed train-test split used for final performance evaluation.

3.1. Representative household selection

To capture the behavioural diversity of electricity consumption within the community while avoiding unnecessary model complexity, this study adopts a contribution-based representative household selection approach. Households are grouped according to their relative contribution to total community electricity demand, computed as the cumulative household consumption expressed as a percentage of the community-wide total over the full observation period. Based on this metric, households are partitioned into six contribution-based bins: (a) below 0.05%, (b) 0.05–0.25%, (c) 0.25–0.5%, (d) 0.5–1%, (e) 1–3%, and (f) greater than 3%, reflecting increasing levels of impact on aggregate demand. From each bin, one representative household is selected based on proximity to the median contribution, ensuring that consumption patterns ranging from very low to

Table 1
Comparative overview of key energy forecasting studies in RECs and buildings.

Study	Building / Community Type	Load Type	Prediction Horizon	Proposed Model(s)	Comparison Models	Key Findings	Limitations
[14]	Residential Community	Electricity Demand	Short-term	XGBoost + SHAP	Classical ML	Improves interpretability, identifies feature importance	Focuses on explainability, not temporal dependencies
[15]	Smart Grid	Electricity Demand	Short-term	RNN–LSTM	RNN, LSTM, ARIMA	Achieved high accuracy (R^2 = 0.9901, RMSE = 0.0187); effectively captured temporal demand patterns	Not REC-specific; lacks deployment-level validation
[16]	Smart Grid	Electricity Demand	Short-term	CNN–LSTM	LSTM, GRU	Hybrid deep learning improved forecasting under volatile demand; scalable for short-term use	Focused on grid-level forecasting; no user-level analysis or interpretability
[17]	Residential and City	Electricity Demand	Day-ahead / Week-ahead	Sparse Transformer	LSTM, RNN-based SOTA	Comparable accuracy with 5x faster inference	Quadratic memory cost for long sequences
[18]	Commercial and Residential Buildings	Electricity Load	Short-term / Multi-horizon	Hybrid Transformer–Liquid Neural Network	LSTM, CNN, GRU	Improved accuracy and training efficiency; robust across datasets	Increased complexity; limited interpretability
[19]	RECs	Aggregate Load	Day-ahead	Bi-LSTM – XGBoost	LSTM, XGBoost, Standard Profiles	Hybrid improves peak and general forecast accuracy	Separate training for components; no real-time test
[23]	Academic Buildings	HVAC & Multi-zone Loads	Multi-horizon	CNN–LSTM–MLP	LSTM, BiLSTM, GRU, CNN	Hybrid reduces error, low memory use, fast training	Limited to academic settings
[24]	Individual Households	Smart Meter Load	Short-term	Global Probabilistic Forecasting	Household baselines	Outperforms point forecasts under concept drift; low data need	Limited interpretability; no hybrid integration
[26]	Community Buildings	Total Load	Short-term	Federated Deep Learning + Adaptive Framework	Centralized DL Models	Privacy-preserving, accurate forecasting	Non-IID data challenges
[27]	Local Energy Communities	Electricity Demand	Short-term	Statistical Baselines (MA, Holt-Winters, Prophet, Lasso, KNN)	Statistical Methods	Forecast accuracy increases with community size	Limited scalability to mixed-use RECs
[28]	Residential Community	Electricity Demand	Short-term	Probabilistic XGBoost	XGBoost, RF	Provides uncertainty estimates with interpretable SHAP analysis	Limited long-term dependency modeling
[29]	Residential Community	Electricity Demand	Short-term	Graph-based Diffusion Probabilistic Model	LSTM, CNN, ARIMA	Provides calibrated uncertainty intervals; robust under volatility	High computational cost; complex architecture
[30]	Transactive Energy Communities	Net Energy Demand	Short-term	Federated Learning Framework	Centralized DL	Preserves privacy; scalable across prosumers	Communication overhead
[31]	Community Microgrids	Electricity Load	Short-term	Meta-heuristic Federated Aggregation (PSO, GWO, DE)	FedAvg	Improved accuracy under heterogeneous data; faster convergence	Higher computational cost; limited field validation

very high demand are explicitly represented in the modelling process. Fig. 2 visualises this selection, with each subplot (a–f) illustrating the contribution distribution within a bin and highlighting the selected representative. A bin-wise sanity check confirms that the selected households lie close to the median behaviour of their respective contribution classes (see Appendix A, Table A.6).

While clustering-based and random sampling strategies could also be employed for household selection [20–22], the objective of the proposed selection approach is to provide transparent and deployment-feasible coverage of low-, medium-, and high-demand regimes rather than to establish a statistically optimal sampling method. Clustering-based methods remain valid alternatives for representative household selection, but they may require parameter tuning, can introduce additional computational overhead, and may reduce the direct traceability of representative assignment in decentralised REC settings. In this study, the contribution-based binning approach provides a simple, reproducible, and operationally interpretable basis for representative household selection. In contrast, the proposed binning strategy is simple, reproducible, and scalable, facilitating interpretability and practical deployment. The selection of six households is therefore a deliberate design choice rather than a data reduction step. It enables the analysis of representative demand archetypes relevant for variability assessment, interpretability, and operational control, while community-level forecasting continues to utilise all households, thereby preserving complete demand dynamics. A quantitative comparison with random sampling and clustering-based approaches is beyond the scope of this work and is identified as an important direction for future research.

3.2. Data preprocessing and feature engineering

At both the household and community levels, data preprocessing was performed to ensure reliability, temporal consistency, and suitability

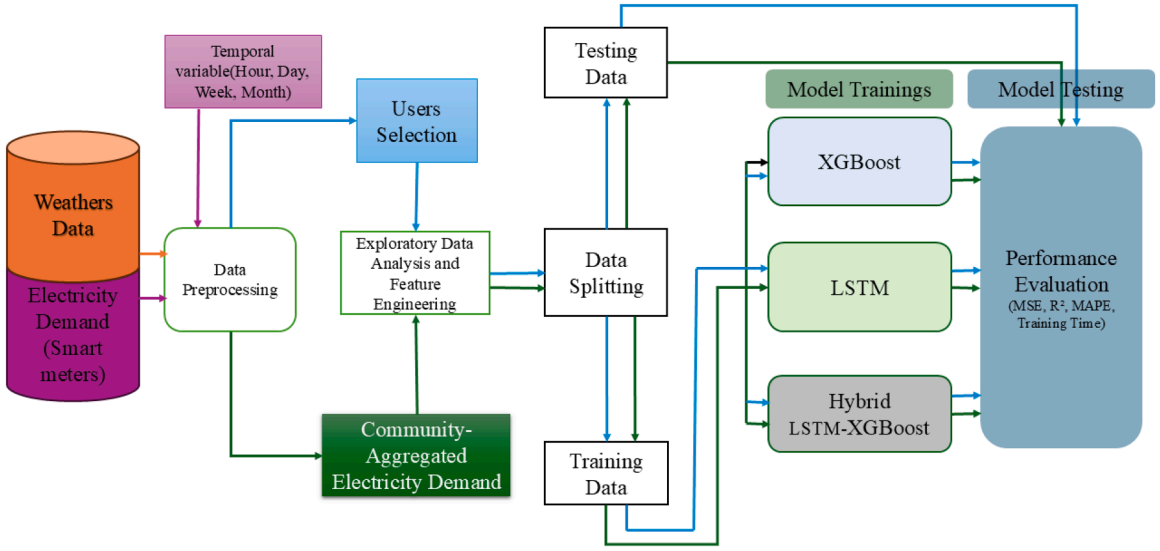


Fig. 1. Proposed research methodology framework. Blue arrows represent household-level forecasting, while green arrows represent community-level forecasting in subsequent steps.

for short-term load forecasting. The raw dataset comprised 15-min electricity consumption measurements from 172 households, spanning the period from May 5, 2022, to September 2, 2023, together with meteorological data at the same temporal resolution. It has been observed that missing smart meter recordings, rather than actual occurrences of zero consumption, are responsible for the zero and missing numbers found in the raw electricity demand data. Households with more than 75% missing or zero-valued measurements were excluded from the analysis to ensure data quality. In addition, time steps in which more than 80% of the household meters contained missing or zero values were removed. After data cleaning and quality control, the final dataset consisted of 107 households and 40,887 valid 15-min observations used consistently across all experiments. After row-level data cleaning and filtering, the processed demand dataset is free of zero-valued observations. Household selection based on the contribution-based binning strategy was performed prior to detailed data cleaning. The selected households were subsequently subjected to the same data quality control and preprocessing procedures, including missing-value imputation and outlier handling, to ensure methodological consistency across all household- and community-level analyses. Weather and electricity consumption data were temporally aligned using a common timestamp at 15-min resolution. Missing meteorological values, including average temperature, relative humidity, and global solar radiation, were imputed using column-wise mean values. Missing electricity consumption values were imputed using a forward-fill followed by a backward-fill strategy to preserve temporal continuity. Outliers in both weather and demand variables were identified using the interquartile range (IQR) method and capped within statistically acceptable bounds to reduce distortion while maintaining the underlying data distribution.

Following preprocessing, aggregate community-level demand was computed by summing the cleaned household-level consumption at each time step. Feature engineering was then applied to capture temporal and behavioral patterns, including calendar-based variables lagged demand features, and short-term rolling averages. Meteorological variables were included to account for weather-sensitive demand effects. For model training and evaluation, a time-aware split was employed to reflect real-world deployment conditions. The first 80% of the time series, covering May 2022 to mid-June 2023, was used for training, while the remaining 20%, from mid-June 2023 to September 2023, was reserved for testing. All input features were standardised using a StandardScaler fitted exclusively on the training data, and the same scaling parameters were subsequently applied to the test set to avoid information leakage. In addition to this fixed split, model robustness under evolving demand conditions was assessed using a month-wise incremental validation scheme. In this setting, models were retrained using all available data up to a given month and evaluated on the subsequent month. This process was repeated from June 2022 to August 2023 to simulate realistic operational conditions in REC applications.

3.3. Model development

This study assesses three architectures: LSTM, XGBoost, and a hybrid LSTM-XGBoost model, in order to provide accurate and intelligible short-term electricity demand forecasting across multiple smart meters. These models were chosen because of how well they complemented one another: XGBoost represents complex nonlinear connections, LSTM captures sequential dependencies, and their hybrid integration seeks to take advantage of both temporal and feature-based learning for RECs. To enable fair performance comparison across forecasting tasks at the community and household levels, each model was trained and validated under identical conditions.

3.3.1. Long short-term memory (LSTM)

LSTM networks are a variant of Recurrent Neural Networks (RNNs) capable of modeling long-range temporal dependencies through internal gating mechanisms that regulate information flow and mitigate vanishing gradient issues [33]. These properties make them particularly effective for high-resolution electricity demand forecasting, where time correlations and nonstationary patterns are dominant. Given an input sequence $\mathbf{x}_t \in \mathbb{R}^p$, the network updates its hidden and cell states as follows:

$$\mathbf{f}_t = \sigma\left(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f\right), \quad (1)$$

$$\mathbf{i}_t = \sigma\left(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i\right), \quad (2)$$

$$\tilde{\mathbf{c}}_t = \tanh\left(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c\right), \quad (3)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (4)$$

$$\mathbf{o}_t = \sigma\left(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o\right), \quad (5)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \quad (6)$$

where $\mathbf{h}_t$ and $\mathbf{c}_t$ denote the hidden and cell states, and $\sigma(\cdot)$ and $\tanh(\cdot)$ are the activation functions. The matrices $\mathbf{W}_{(\cdot)}$, $\mathbf{U}_{(\cdot)}$, and $\mathbf{b}_{(\cdot)}$ are trainable parameters. To project the sequential representation into a latent feature

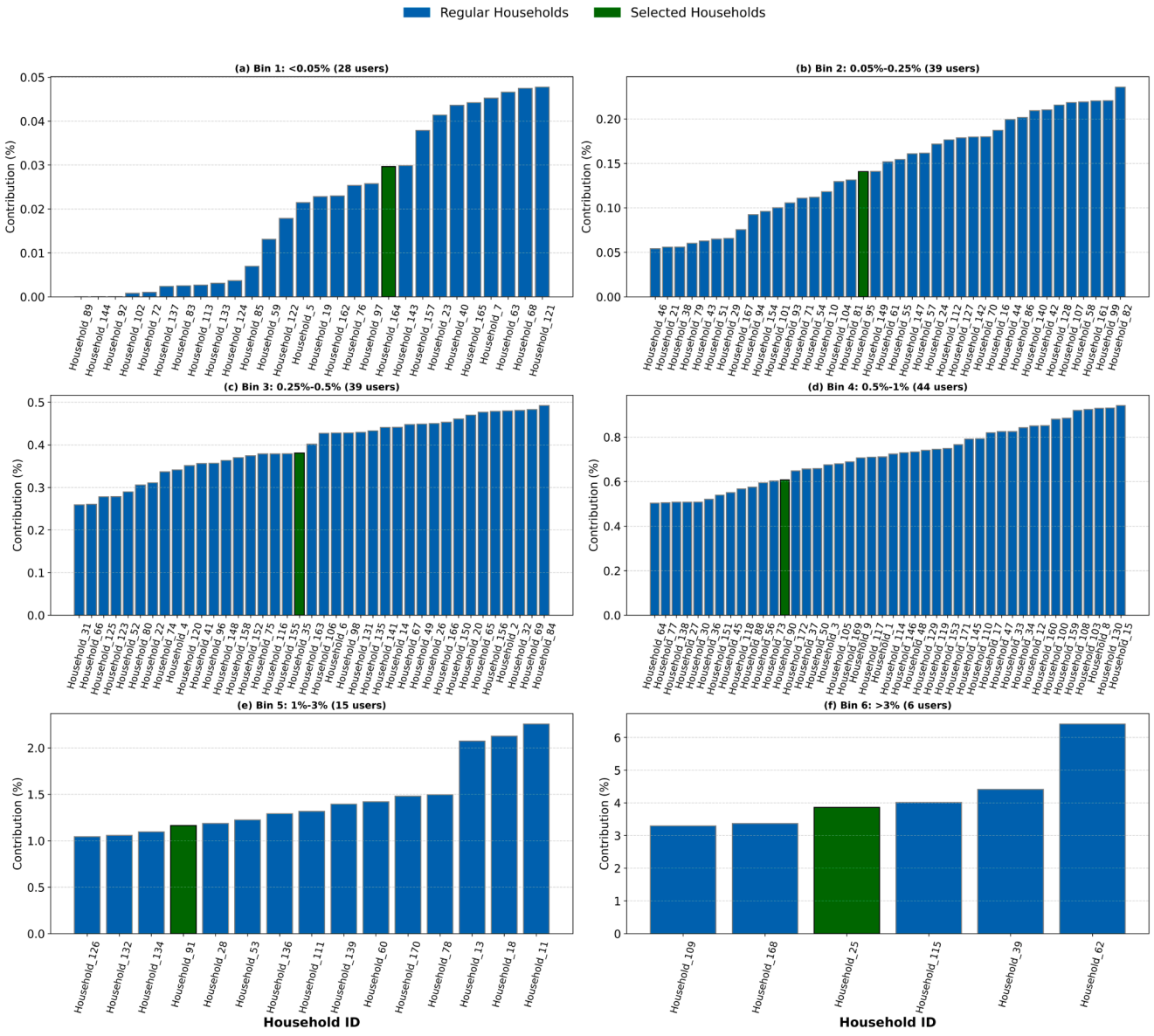


Fig. 2. Households grouped into contribution-based bins, with one randomly selected household highlighted in each bin.

space, the hidden state is passed through a dense layer:

$$\mathbf{z}_t = \phi_\theta(\mathbf{x}_t) = \mathrm{ReLU}\big(\mathbf{W}_d \mathbf{h}_t + \mathbf{b}_d\big), \tag{7}$$

where ϕ_θ denotes the LSTM encoder parameterized by θ. The model output for time t is obtained as:

$$\hat{y}_t^{(\mathrm{LSTM})} = \mathbf{w}_y^\top \mathbf{z}_t + b_y. \tag{8}$$

The model minimizes mean squared error (MSE):

$$\mathcal{L}_{\mathrm{LSTM}} = \frac{1}{N}\sum_{t=1}^{N}\left(y_t - \hat{y}_t^{(\mathrm{LSTM})}\right)^2. \tag{9}$$

In this framework, the LSTM encoder serves as a sequence feature extractor that captures both short-term and long-term demand dependencies, which are essential for downstream hybrid modeling.

3.3.2. Extreme gradient boosting (XGBoost)

XGBoost is an ensemble-based gradient boosting algorithm that builds additive regression trees to capture nonlinear feature interactions [34] efficiently. For a given input $\mathbf{x}_t$, the prediction is expressed as:

$$\hat{y}_t = \sum_{k=1}^{K} f_k(\mathbf{x}_t), \quad f_k \in \mathcal{F}, \tag{10}$$

where $\mathcal{F}$ denotes the functional space of regression trees $\{f_k\}_{k=1}^{K}$. Each tree f_k is trained sequentially to minimize the regularized objective:

$$J^{(k)} = \sum_{t=1}^{N} \ell\left(y_t, \hat{y}_t^{(k-1)} + f_k(\mathbf{x}_t)\right) + \Omega(f_k), \tag{11}$$

where $\Omega(f_k)$ penalizes tree complexity to prevent overfitting. XGBoost's ability to capture nonlinear dependencies and quantify feature importance makes it a strong candidate for modeling the feature-level relationships in smart meter data.

3.3.3. Hybrid LSTM–XGBoost architecture

A sequential hybrid forecasting framework is proposed to address the nonlinear and highly variable nature of residential electricity demand. The framework integrates a LSTM network with a gradient-boosted tree regressor through a two-stage learning strategy, combining the strengths

of deep sequence modelling and tree-based nonlinear regression (Algorithm 1).

Algorithm 1: Training the LSTM→XGBoost hybrid.

Input: Chronological dataset $\{(\mathbf{x}_t, y_t)\}_{t=1}^{T}$; train split ratio α (0.8)
Output: Trained LSTM encoder $\phi_{\hat{\theta}}$ and XGBoost regressor $F(\cdot)$
Feature engineering: Build $\mathbf{x}_t$ (weather, calendar one-hots, lags, rolling means). Set $\mathrm{Lag}_1 = \mathrm{shift}(1)$, $\mathrm{Lag}_{24} = \mathrm{shift}(24)$ (not 2), and causal rolling mean.;
Chronological split: $\mathcal{D}_{\text{train}} = \{t \le \lfloor \alpha T \rfloor\}$, $\mathcal{D}_{\text{test}} = \{t > \lfloor \alpha T \rfloor\}$.;
Standardization (fit-on-train only): Compute $\boldsymbol{\mu}, \boldsymbol{\sigma}$ on $\mathcal{D}_{\text{train}}$; $\tilde{\mathbf{x}}_t = (\mathbf{x}_t - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$.;
Phase A: LSTM encoder training: Build LSTM encoder ϕ_θ with a temporary regression head $\hat{y}_t^{(\mathrm{LSTM})} = \mathbf{w}_y^\top \phi_\theta(\tilde{\mathbf{x}}_t) + b_y$.;
Minimize $\frac{1}{N}\sum_{t\in\mathcal{D}_{\text{train}}}(y_t - \hat{y}_t^{(\mathrm{LSTM})})^2$ with Adam + early stopping.;
After convergence, discard the head; keep $\phi_{\hat{\theta}}$.;
Feature extraction: For all t, compute latent features $\mathbf{z}_t = \phi_{\hat{\theta}}(\tilde{\mathbf{x}}_t)$.;
Phase B: Boosting on latent features: Train XGBoost $F(\cdot)$ on pairs $(\mathbf{z}_t, y_t)$ for $t \in \mathcal{D}_{\text{train}}$ using the regularized second-order objective.;
Validation: Predict $\hat{y}_t = F(\mathbf{z}_t)$ on $t \in \mathcal{D}_{\text{test}}$ and report MSE, MAPE, R^2.;
return $\{\boldsymbol{\mu}, \boldsymbol{\sigma}\}$, *encoder* $\phi_{\hat{\theta}}$, *and booster* $F(\cdot)$

In the first stage, an LSTM network is employed to capture temporal dependencies in historical electricity consumption. Given a sequence of past load observations $\mathbf{x}_t$, the LSTM encodes temporal consumption patterns into a latent representation $\mathbf{z}_t = \phi_\theta(\mathbf{x}_t)$, where $\phi_\theta(\cdot)$ denotes the LSTM mapping parameterised by θ. These latent representations summarise sequential dynamics that are difficult to model using conventional regression techniques (Algorithm 2).

Algorithm 2: Inference with the LSTM→XGBoost hybrid.

Input: New feature vector $\mathbf{x}_\tau$
Output: Prediction $\hat{y}_\tau$
Load: $\{\boldsymbol{\mu}, \boldsymbol{\sigma}\}$, trained encoder $\phi_{\hat{\theta}}$, trained booster $F(\cdot)$.;
Preprocess: Standardize $\tilde{\mathbf{x}}_\tau = (\mathbf{x}_\tau - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$; Ensure the required lags/rolling statistics are available.;
Encode: $\mathbf{z}_\tau = \phi_{\hat{\theta}}(\tilde{\mathbf{x}}_\tau)$.;
Predict: $\hat{y}_\tau = F(\mathbf{z}_\tau)$.;
return $\hat{y}_\tau$

In the second stage, the extracted latent representations are concatenated with exogenous variables $\mathbf{v}_t$, including meteorological conditions and calendar-related information. This enriched feature vector is subsequently provided as input to an XGBoost regressor, which exploits the learned latent feature space to model complex nonlinear dependencies in electricity demand. The resulting prediction function is defined as:

$$\hat{y}_t = F\big(\phi_\theta(\mathbf{x}_t), \mathbf{v}_t\big), \tag{12}$$

where $F(\cdot)$ denotes the XGBoost ensemble composed of K regression trees:

$$F(\cdot) = \sum_{k=1}^{K} f_k(\cdot). \tag{13}$$

Model training is performed sequentially rather than through joint or end-to-end optimisation. The LSTM component is first trained independently to learn temporal representations from historical load sequences. Once trained, the resulting latent representations are fixed and subsequently used as input features for training the XGBoost regressor. This two-stage strategy enhances model transparency and reproducibility while allowing each component to focus on the task it is best suited for. Since the goal of the LSTM component is to function as a temporal feature encoder rather than a stand-alone predictor, the regression head of the LSTM is purposefully eliminated in the hybrid arrangement. This design decision enables the ensuing XGBoost model to take advantage of both exogenous variables and latent representations created from the LSTM inside an open, tree-based regression framework. Due to minimal performance advantages proportional to the increased training complexity and decreased interpretability, two important factors for implementation in REC applications joint or end-to-end optimisation was not pursued. To ensure a fair and consistent comparison across models, the LSTM and XGBoost components are tuned independently using grid search over their respective hyperparameter spaces. For the hybrid configuration, the optimal hyperparameters identified for each component are adopted without joint optimisation. The selected hyperparameter values and corresponding search ranges are reported in Table 2, ensuring consistency across all model configurations during training and evaluation. Based on the empirical distribution of residuals computed on a calibration subset of the training data, prediction ranges were constructed using a conformal prediction approach applied to the hybrid model outputs in order to account for forecast uncertainty (Fig. 3).

3.4. Evaluation metrics

To quantitatively evaluate forecasting accuracy across different model configurations and scales, three widely adopted performance metrics were used. Mean Absolute Error (MAE), defined in Eq. (14), measures the average magnitude of forecast errors, expressed in the same units as the target variable. Mean Absolute Percentage Error (MAPE) Eq. (15) expresses forecast error as a percentage of the actual value. It facilitates comparison across different consumption levels but can be sensitive to near-zero values. The Coefficient of Determination (R^2) metric, shown in Eq. (16), quantifies the proportion of variance in the actual demand that the model captures.

$$\mathrm{MAE} = \frac{1}{N}\sum_{t=1}^{N}|y_t - \hat{y}_t| \tag{14}$$

$$\mathrm{MAPE} = \frac{100\%}{N}\sum_{t=1}^{N}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{15}$$

$$R^2 = 1 - \frac{\sum_{t=1}^{N}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{N}(y_t - \bar{y})^2} \tag{16}$$

All metrics were computed separately at both the household and community levels. Lower MAE and MAPE values, along with higher R^2, indicate superior model performance. These evaluation criteria enable rigorous and interpretable comparisons between standalone and hybrid forecasting models across varying user behaviours and aggregation levels.

3.5. SHAP for model interpretability

To enhance the transparency of the forecasting models, Shapley Additive exPlanations (SHAP), a widely used game-theoretic approach for feature attribution, is employed. SHAP values originate from cooperative game theory and assign each feature a contribution score that reflects its marginal impact on the final model prediction. Unlike traditional sensitivity analysis or coefficient-based measures, SHAP provides both global interpretability, by ranking features according to their average contribution across the dataset, and local interpretability, by explaining individual predictions in terms of the effects of each feature. This dual perspective allows SHAP to capture not only the relative importance of input variables, but also the direction of their influence

Table 2
Optimal hyperparameters of LSTM, XGBoost, and Hybrid (LSTM-XGBoost).

Method	Hyper-parameters	Search space of hyper-parameters	Optimal values
LSTM	Number of hidden units (LSTM)	[32, 64, 128]	64
	Dropout rate	[0.1, 0.2, 0.3, 0.5]	0.3
	Number of hidden units (Dense)	[16, 32, 64]	32
	Optimizer	[Adam, RMSprop, SGD]	Adam
	Epochs / Batch size	[20–100], [16, 32, 64]	50 / 32
XGBoost	Number of estimators	[50, 100, 200, 500]	100
	Learning rate	[0.01, 0.05, 0.1, 0.2]	0.1
	Maximum depth	[2, 4, 6, 8]	4
	Random state / Verbosity	[42], [0,1]	42 / 0
LSTM-XGBoost	LSTM hidden units	[32, 64, 128]	64
	Dropout rate	[0.1, 0.2, 0.3, 0.5]	0.3
	Dense layers in LSTM	[16, 32, 64]	32 + 16
	LSTM optimizer/loss	[Adam, RMSprop], [MSE, MAE]	Adam / MSE
	Epochs / Batch size	[20–100], [16, 32, 64]	50 / 32
	XGBoost estimators	[50, 100, 200, 500]	100
	XGBoost learning rate / max depth	[0.01–0.2], [2, 4, 6, 8]	0.1 / 4

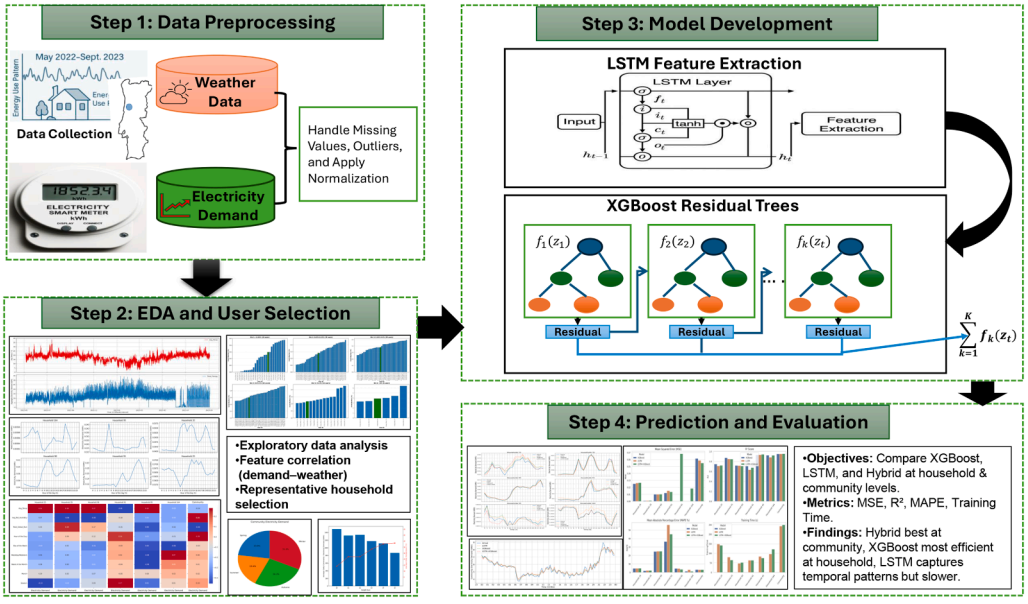


Fig. 3. Two-stage LSTM–XGBoost hybrid model with sequential training, where LSTM-extracted features are used for XGBoost-based forecasting.

(positive or negative) on the target outcome. In the context of electricity demand forecasting, SHAP enables the decomposition of model outputs into interpretable components, clarifying whether weather conditions, temporal patterns, or recent consumption history are driving changes in predicted demand. This improves transparency and supports stakeholder trust in the deployment of advanced ML models within RECs.

4. Datasets exploration and analysis

Exploratory Data Analysis (EDA) plays a crucial role in understanding the characteristics and structure of the dataset before developing predictive models. This section examines the distributions, trends, and underlying patterns of weather and electricity demand. Through visual interpretation and statistical summaries, EDA helps identify potential deviations, user variability, and seasonality. These insights are crucial for ensuring the robustness of forecasting and guiding model development. The descriptive statistics presented in Table 3 are provided to illustrate the variability of electricity consumption patterns across distinct contribution classes within the community. The selected households are not intended to replicate population-level average demand, but rather to reflect representative demand regimes spanning low, intermediate, and high consumption levels. Accordingly, the household-level analysis is designed to examine model behaviour under heterogeneous consumption conditions, rather than to deliver a statistical characterisation of the entire household population. These statistics summarise the distribution and variability of key variables over the whole observation period. Table 3 reports the key summary metrics, including the mean, standard deviation, quartiles, and range, for selected weather attributes, six representative households, and the total community demand.

The Count column indicates the total number of valid observations available for each variable. Smart meter datasets may contain missing observations due to intermittent communication failures, temporary device malfunctions, or data transmission interruptions. Such incomplete records were excluded during data preprocessing; consequently, all descriptive statistics reported in Table 3 are derived from the cleaned dataset. The 25%, 50%, and 75% columns represent the first quartile (Q1), median (Q2), and third quartile (Q3), respectively. These fluctuations are expected to strongly influence demand patterns, as evident in the higher temperatures during the summer and lower temperatures during the winter. Descriptive statistics for solar radiation were calculated using only daylight-period values, with nighttime observations excluded due to their physical nullity and potential to skew distributional summaries. The dataset enables detailed anal-

Table 3
Descriptive statistics of selected weather variables, six representative households, and total community electricity demand (May 5 2022 – September 2 2023).

Variable (Unit)	Count	Mean	Std	Min	25%	50%	75%	Max
Average Temperature (°C)	40887	17.060	4.349	1.600	14.400	17.350	19.900	38.400
Relative Humidity (%)	40887	78.951	14.431	19.500	71.000	82.000	90.500	100.000
Solar Radiation (W/m^2)	26459	216.958	180.134	0.050	46.400	180.000	376.700	692.600
Household 25 (kWh)	40887	0.564	0.088	0.244	0.466	0.582	0.614	0.836
Household 35 (kWh)	40887	0.046	0.004	0.031	0.043	0.045	0.047	0.063
Household 90 (kWh)	40887	0.049	0.023	0.000	0.030	0.048	0.063	0.127
Household 91 (kWh)	40887	0.086	0.072	0.000	0.041	0.073	0.092	0.407
Household 95 (kWh)	40887	0.004	0.001	0.003	0.004	0.004	0.004	0.006
Household 164 (kWh)	40887	0.004	0.001	0.002	0.003	0.004	0.004	0.006
Community Electricity Demand (kWh)	40887	6.075	1.210	0.466	5.182	5.830	6.723	9.988

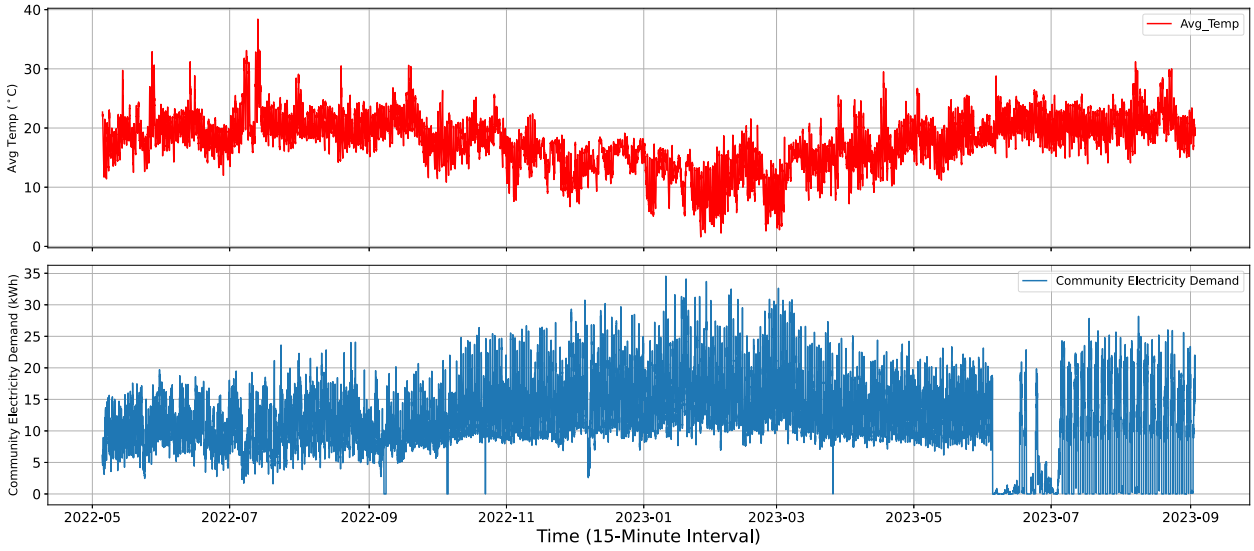


Fig. 4. Time-series of average temperature (top) and total community electricity demand (bottom), recorded at 15-min intervals from May 2022 to September 2023.

ysis of short-term dynamics, forming a robust basis for the correlation analysis and feature engineering described in the subsequent sections.

4.1. Spatiotemporal analysis of electricity demand

Fifteen-minute interval records of ambient temperature and community electricity demand were analysed for the period from May 2022 to September 2023. In the Fig. 4, community demand peaks in winter, primarily due to heating and lighting requirements, and declines in summer months. Behavioural effects such as appliance usage, occupancy schedules, and efficiency measures introduce short-term fluctuations superimposed on these seasonal cycles.

These observations confirm that both climatic and behavioural factors influence short-term load variability. Temperature is therefore retained as a key exogenous predictor in the hybrid forecasting model to capture seasonal and diurnal demand dynamics effectively. Hourly average electricity demand profiles were derived for six representative households to capture intra-day consumption dynamics. Fig. 5 illustrates the 24-hour mean demand (kWh) for each household, averaged over the whole observation period from May 2022 to September 2023. Distinct behavioural patterns emerge across users. Household 25 maintains the highest and most consistent demand, with midday and afternoon peaks indicative of continuous appliance usage. Household 164 shows an almost flat, near-zero profile, reflecting minimal occupancy. Household 90 exhibits a bimodal pattern with morning and evening peaks, whereas Household 91 demonstrates a gradual increase throughout the day, culminating in an evening maximum typical of post-work routines. Households 35 and 95 record moderate fluctuations with relatively low consumption. These differentiated load profiles reveal substantial heterogeneity in daily routines, underscoring the need for user-level modeling in RECs. Capturing such intra-day diversity justifies the inclusion of temporal features (hour, day, and season) in the forecasting framework to enhance prediction accuracy and interpretability.

Electricity demand and daylight duration were compared across four meteorological seasons, Spring (March 21–June 20), Summer (June 21–September 22), Autumn (September 23–December 20), and Winter (December 21–March 20). At the community level, demand peaks in winter (31.4 %) and reaches its minimum in summer (19.4 %), as shown in Fig. 6. This behaviour aligns with climatic variation and daylight availability, which range annually from approximately 9.4 to 15.1 h of daylight per day. Daylight duration was estimated from 15-min solar radiation data using a threshold-based detection method, a reliable approach widely adopted in solar energy studies.

Extended daylight periods coincide with higher temperatures and reduced electricity usage, reflecting lower heating and lighting requirements. Conversely, shorter winter days correspond to increased demand driven by heating and illumination needs.

4.2. Correlation analysis

Pearson correlation analysis was conducted to explore linear relationships between electricity demand and selected meteorological and temporal variables at both the community and household levels. The resulting correlation matrix is visualized in Fig. 7. Correlation patterns vary considerably across users, reflecting behavioural diversity within the community. Household 25 shows a mild positive correlation with temperature suggesting that higher temperatures correspond to slightly

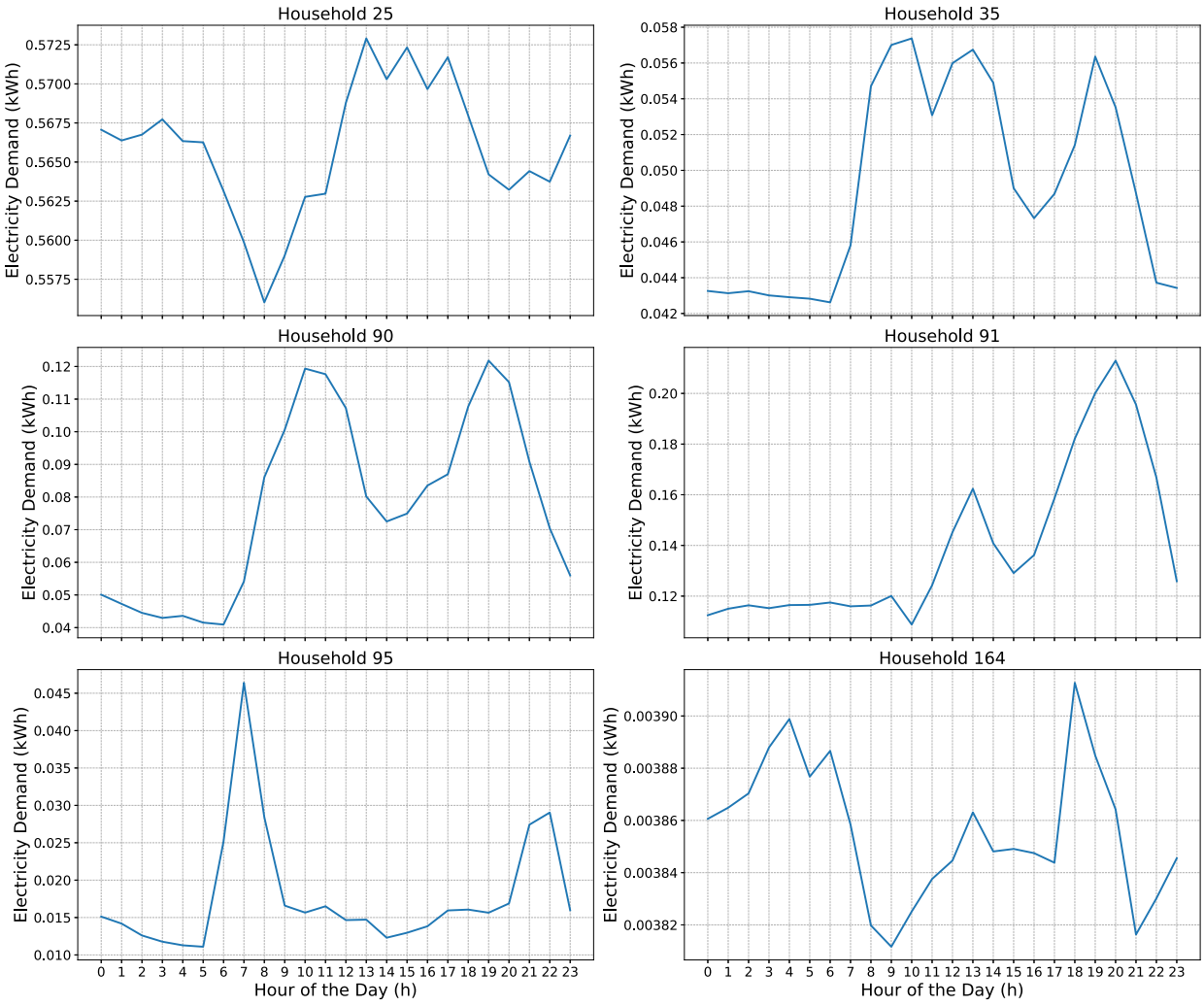
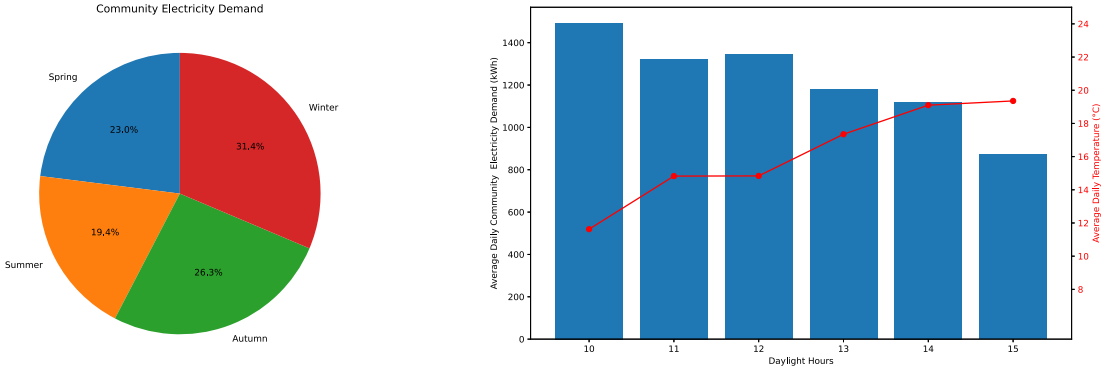


Fig. 5. Average hourly electricity demand profiles for six representative households.



(a) Seasonal distribution of community electricity demand.

(b) Effect of daylight hours on energy use and temperature.

Fig. 6. Combined seasonal and daylight-based community energy analysis.

increased usage, possibly due to the use of cooling devices. Household 90 displays a strong negative correlation with temperature, while Household 164 exhibits moderate negative relationships with both humidity and temperature, and weak positive associations with solar radiation and seasonal index. Such differences indicate distinct occupant behaviours and varying appliance dependencies across households. At the community level, electricity demand correlates moderately and positively with temperature, consistent with the cooling-related loads that occur during warmer periods. Weak associations are observed with solar radiation and humidity. Temporal indicators (hour, day, and season) show low correlation values, while month exhibits a slight negative relationship. These findings confirm that environmental factors influence community-level demand, whereas household-level responses remain highly individualized. Such heterogeneous relationships justify

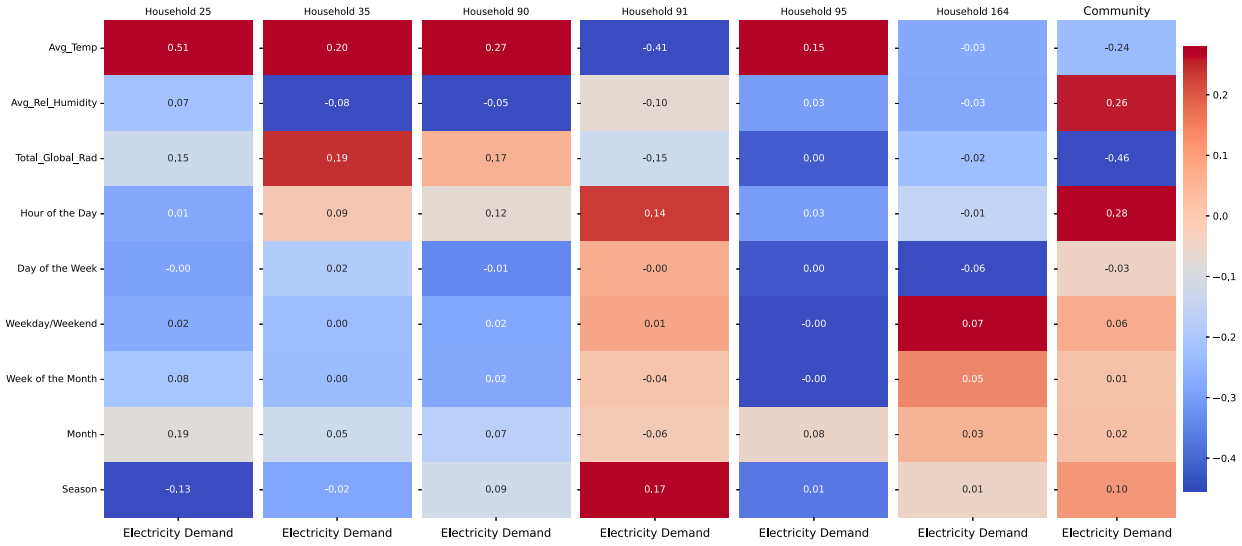


Fig. 7. Pearson correlation heatmap between electricity demand and selected environmental and temporal variables for the community and six representative households.

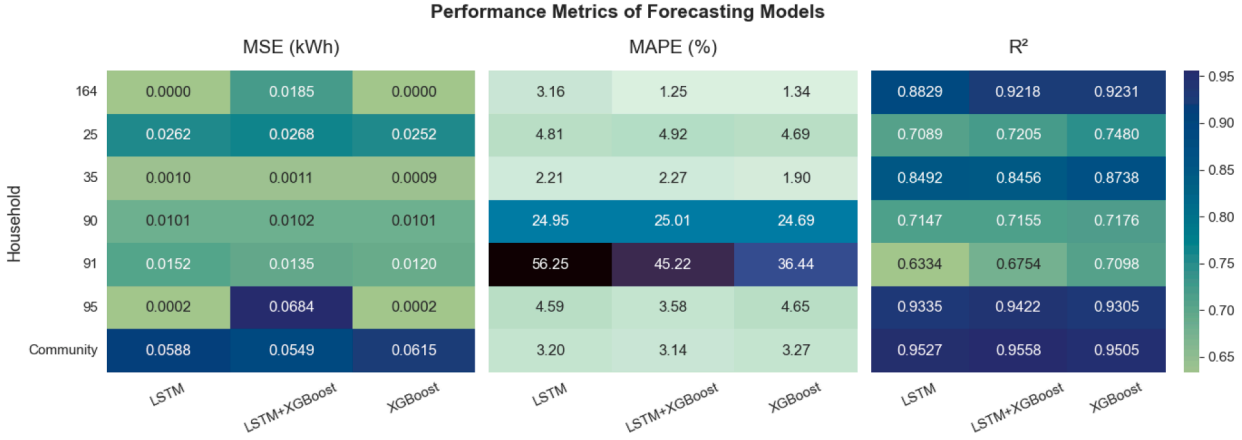


Fig. 8. Comparison of LSTM, XGBoost, and hybrid models using MSE, MAPE, and R^2 for representative households and community.

a disaggregated, multi-level approach to forecasting in RECs. By integrating both meteorological and temporal features within the hybrid LSTM-XGBoost framework, the model effectively captures nonlinear, user-specific dependencies that purely aggregate or linear models might overlook. The exploratory analyses collectively reveal that diverse behavioural, temporal, and climatic factors shape electricity demand in the studied REC. High variability across households and seasons underscores the necessity of user-level modelling, while the moderate correlations with weather variables confirm their importance as exogenous predictors. These insights directly inform the hybrid LSTM-XGBoost forecasting framework developed in the next section, guiding the selection of temporal and meteorological features and validating the multi-level design used for both household and community-level prediction.

5. Results

The proposed forecasting framework was evaluated using 15-min interval data from a REC located in Loureiro, Portugal (Section 3). Two forecasting scales were considered: *(i)* the community level, based on aggregated household demand, and *(ii)* the household level, focusing on six households randomly sampled within predefined contribution-based bins to illustrate heterogeneous demand regimes. Three predictive models were tested: *(a)* a standalone LSTM model capturing sequential dependencies, *(b)* an XGBoost regressor modelling nonlinear feature interactions, and *(c)* a hybrid LSTM-XGBoost model combining both strengths through sequential encoding and structured regression. All models used identical input features, including meteorological variables (temperature, humidity, solar radiation, wind speed, and direction), temporal indicators (hour, day, month, season), and lag-based demand features (previous values and rolling averages). A month-wise incremental validation strategy simulated real-world deployment, with models trained cumulatively on past months and evaluated on the subsequent month.

5.1. Evaluation of predictive accuracy and model comparison

Table 4 and Fig. 8 summarise forecasting accuracy and training time across all configurations. Performance varies by user, reflecting heterogeneous demand behaviour within the REC. The hybrid model showed its clearest gains at the community level and in selected variable-demand cases, while XGBoost remained highly competitive across several households and often achieved similar or lower error with substantially faster training. For relatively stable users such as 35 and 164, XGBoost and the hybrid model produced similarly low errors, with XGBoost offering substantially lower retraining cost. LSTM alone provided reasonable accuracy but required longer computation time, limiting its suitability for frequent retraining. At the community level, all models demonstrated high predictive power, with the hybrid con-

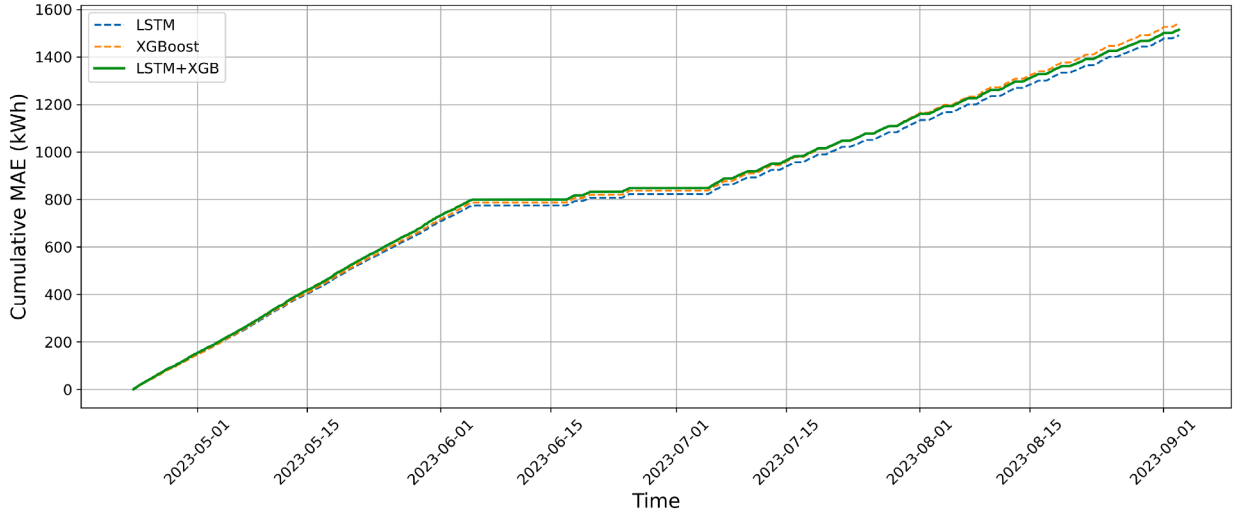


Fig. 9. Cumulative MAE for community-level forecasting using XGBoost, LSTM, and Hybrid LSTM-XGBoost models. The hybrid model maintains the lowest cumulative error, indicating strong robustness.

Table 4
Forecasting accuracy and training time of XGBoost, LSTM, and hybrid models at community and household levels.

	Model	MSE (kWh)	R^2	MAPE (%)	Training Time (s)
Household 25	XGBoost	0.0252	0.7480	4.69	0.41
	LSTM	0.0262	0.7089	4.81	148.63
	LSTM-XGBoost	0.0268	0.7205	4.92	141.73
Household 35	XGBoost	0.0009	0.8738	1.90	0.40
	LSTM	0.0010	0.8492	2.21	80.11
	LSTM-XGBoost	0.0011	0.8456	2.27	64.57
Household 90	XGBoost	0.0101	0.7176	24.69	0.41
	LSTM	0.0101	0.7147	24.95	47.66
	LSTM-XGBoost	0.0102	0.7155	25.01	52.20
Household 91	XGBoost	0.0120	0.7098	36.44	0.39
	LSTM	0.0152	0.6334	56.25	63.39
	LSTM-XGBoost	0.0135	0.6754	45.22	63.44
Household 95	XGBoost	0.0002	0.9305	4.65	0.26
	LSTM	0.0002	0.9335	4.59	83.76
	LSTM-XGBoost	0.0002	0.9422	3.58	75.34
Household 164	XGBoost	0.0000	0.9231	1.34	0.25
	LSTM	0.0000	0.8829	3.16	96.42
	LSTM-XGBoost	0.0000	0.9218	1.25	121.65
Community	XGBoost	0.0615	0.9505	3.27	0.69
	LSTM	0.0588	0.9527	3.20	242.69
	LSTM-XGBoost	0.0549	0.9558	3.14	249.80

figuration yielding the best performance, as indicated by $R^2 = 0.9558$ and MAPE = 3.14%. Beyond point forecasting accuracy, the uncertainty of the proposed approach was evaluated through conformal prediction. For a nominal 90% prediction interval, the empirical coverage on the test set reached 92.6%, indicating that the resulting prediction intervals are well calibrated and suitable for risk-aware decision-making in REC applications.

These findings indicate a trade-off between retraining efficiency, predictive accuracy, and adaptability under different demand regimes. Overall, the hybrid model provides competitive performance across household and community scales, with the clearest gains observed at the community level and under more dynamic demand conditions, while XGBoost offers stronger efficiency and more consistent average month-wise robustness. The monthly distribution of MAPE at the community level is shown in Fig. 11. The observed variation remains moderate across most months, suggesting that the reported average error is not driven by a small subset of favourable periods.

Table 5
Month-wise comparison of MAPE (%) and R^2 scores across forecasting models using incremental learning.

Test Month	MAPE (%)			R^2 Score		
	XGBoost	LSTM	Hybrid	XGBoost	LSTM	Hybrid
2022-06	3.66	18.09	12.75	0.9374	-0.5415	0.1231
2022-07	2.78	4.17	3.22	0.9485	0.8906	0.9227
2022-08	3.18	3.37	3.42	0.9462	0.9340	0.9378
2022-09	3.51	4.58	5.30	0.9406	0.8947	0.8408
2022-10	4.84	6.36	5.58	0.9185	0.8626	0.8877
2022-11	4.10	4.44	4.59	0.9439	0.9284	0.9272
2022-12	3.82	8.31	7.59	0.9307	0.6426	0.7141
2023-01	3.60	4.50	4.28	0.9409	0.9095	0.9152
2023-02	3.78	4.62	4.45	0.9414	0.9126	0.9160
2023-03	3.90	4.47	4.23	0.9405	0.9160	0.9300
2023-04	3.75	3.98	3.97	0.9410	0.9333	0.9320
2023-05	3.70	4.06	3.85	0.9434	0.9333	0.9381
2023-06	3.41	5.88	5.79	0.8928	0.7110	0.7076
2023-07	3.47	3.76	3.58	0.9206	0.8971	0.9081
2023-08	3.06	3.67	3.63	0.9258	0.8801	0.8902
Average	3.67	5.54	4.86	0.9342	0.7803	0.8327

The cumulative MAE results further highlight the hybrid model's robustness to temporal fluctuations. At the community level Fig. 9, it consistently achieves the lowest cumulative error over time, demonstrating resilience to load variability. At the household level, Fig. 10 shows that improvements are most evident for users with irregular or dynamic consumption patterns. The hybrid model thus maintains stability in sequential forecasts, thereby reducing error accumulation, a key factor for reliable scheduling and battery operation in RECs.

5.2. Forecast evaluation and robustness analysis

Model robustness for community level was examined using a month-wise incremental learning strategy to simulate the evolution of REC conditions over time. Each model was retrained cumulatively on all preceding months and tested on the subsequent one, reflecting real-world adaptation to seasonal and behavioural changes. The evaluation covered a 15-month horizon (June 2022–August 2023), encompassing both winter- and summer-dominant periods to assess temporal generalisation. Table 5 reports month-wise MAPE and R^2 values. XGBoost achieved the most consistent performance, with a mean MAPE of 3.67% and R^2 of 0.9341, maintaining stability across nearly all months. The hybrid LSTM-XGBoost model exhibited intermediate performance overall, with an MAPE of 4.86% and an R^2 of 0.8327, respectively.

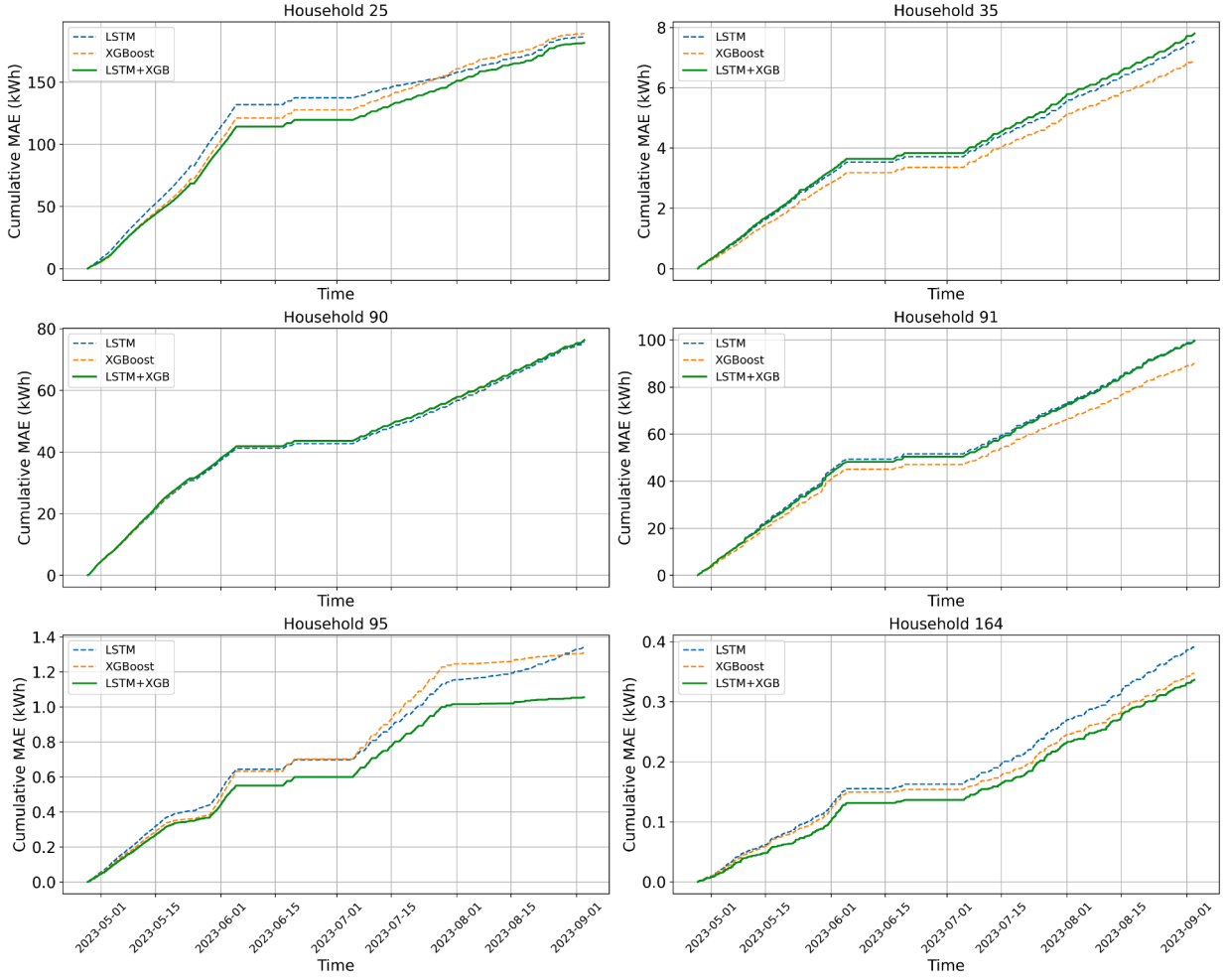


Fig. 10. Cumulative MAE comparison across six representative households. The hybrid model generally achieves lower cumulative error, especially for households with high variability.

However, it showed stronger resilience during volatile periods, such as December 2022 and June 2023, when sudden load fluctuations occurred. The standalone LSTM model performed less reliably, with higher errors and instability, most notably in June 2022, as R^2 showed negative values, highlighting its sensitivity to data drift and its limited adaptability without structured feature support. This behaviour occurs during the first step of the month-wise incremental validation, where the LSTM is trained on a very limited historical window, resulting in a cold-start condition that particularly affects sequence-based models requiring longer temporal context. These results reveal a trade-off between accuracy and adaptability. XGBoost remains the most robust option for stable long-term forecasting, while the hybrid model offers superior resilience under dynamic conditions by combining sequential and feature-based learning. Such robustness is essential for day-ahead scheduling, demand response, and energy storage optimisation in RECs, ensuring reliable operation even during atypical demand events. Fig. 12 shows the comparison between actual and predicted mean daily electricity demand for six households during the testing period. All models capture the general profile, though deviations occur for highly variable users 25 and 91, where the hybrid approach more accurately follows rapid demand transitions. For stable households 35 and 164, all models perform similarly. Fig. 13 presents the hourly forecast for 25 May 2023, showing a close alignment between the predicted and actual community-level electricity demand throughout the day. The model effectively captures both peak and recovery phases. Similarly, Fig. 14 displays the aggregated weekly profile beginning on 12 May 2023, further confirming the model's reliability in maintaining accurate predictions over extended periods for short-term operational forecasting. This shows a trade-off between the average robustness and adaptability when faced with volatile demand conditions, suggesting that the model selection in RECs is dependent on the variability characteristics. In summary, the hybrid LSTM-XGBoost framework consistently provides accurate, stable, and computationally feasible forecasts across both temporal and spatial scales. Its consistent error reduction and adaptability make it a practical solution for real-time energy management and planning in REC.

5.3. *Model interpretability based on SHAP analysis*

Model interpretability is essential for the practical deployment of forecasting frameworks in RECs. Beyond prediction accuracy, understanding how input variables influence model decisions enables transparent and trustworthy operation. SHAP analysis was employed to quantify the contribution of each feature in the hybrid LSTM-XGBoost model at both the household and community levels. The gradient-boosting part of the hybrid system is the subject of interpretability analysis in this paper, while the LSTM functions as a temporal feature encoder. Through lagged and rolling inputs, the LSTM records sequential consumption trends. These inputs are then incorporated into latent representations that the XGBoost regressor uses. Consequently, the significance of LSTM-derived features in the tree-based model implicitly reflects the impact of temporal dependencies. At the household scale, distinct attribution

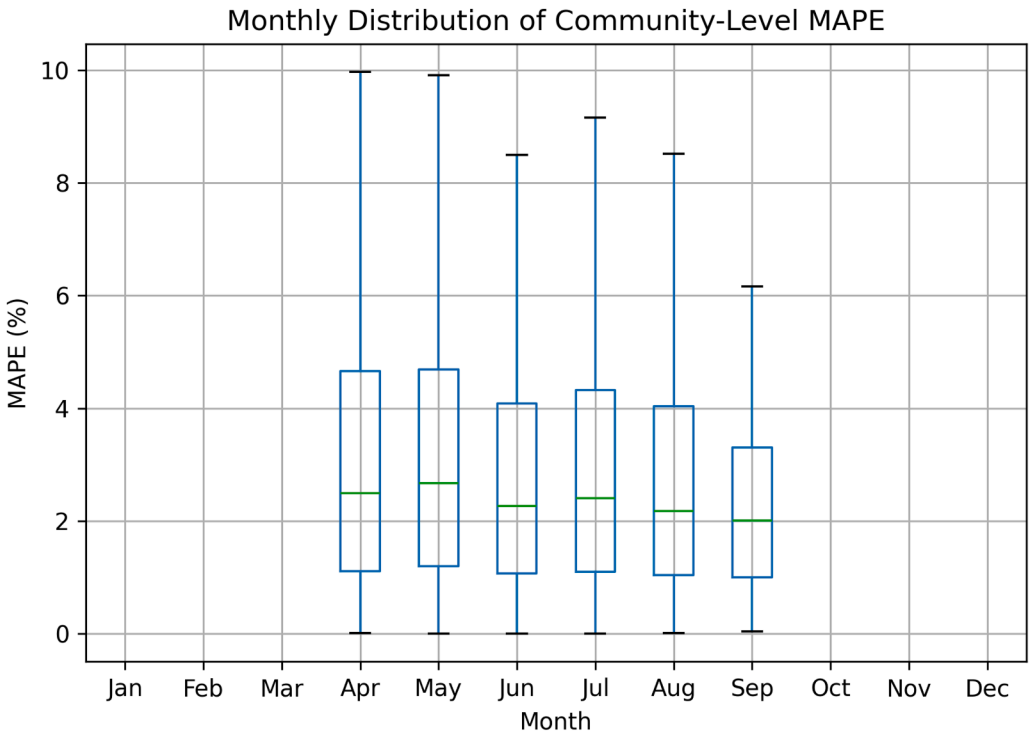


Fig. 11. Monthly distribution of community-level mean absolute percentage error (MAPE) obtained from the proposed hybrid LSTM–XGBoost model.

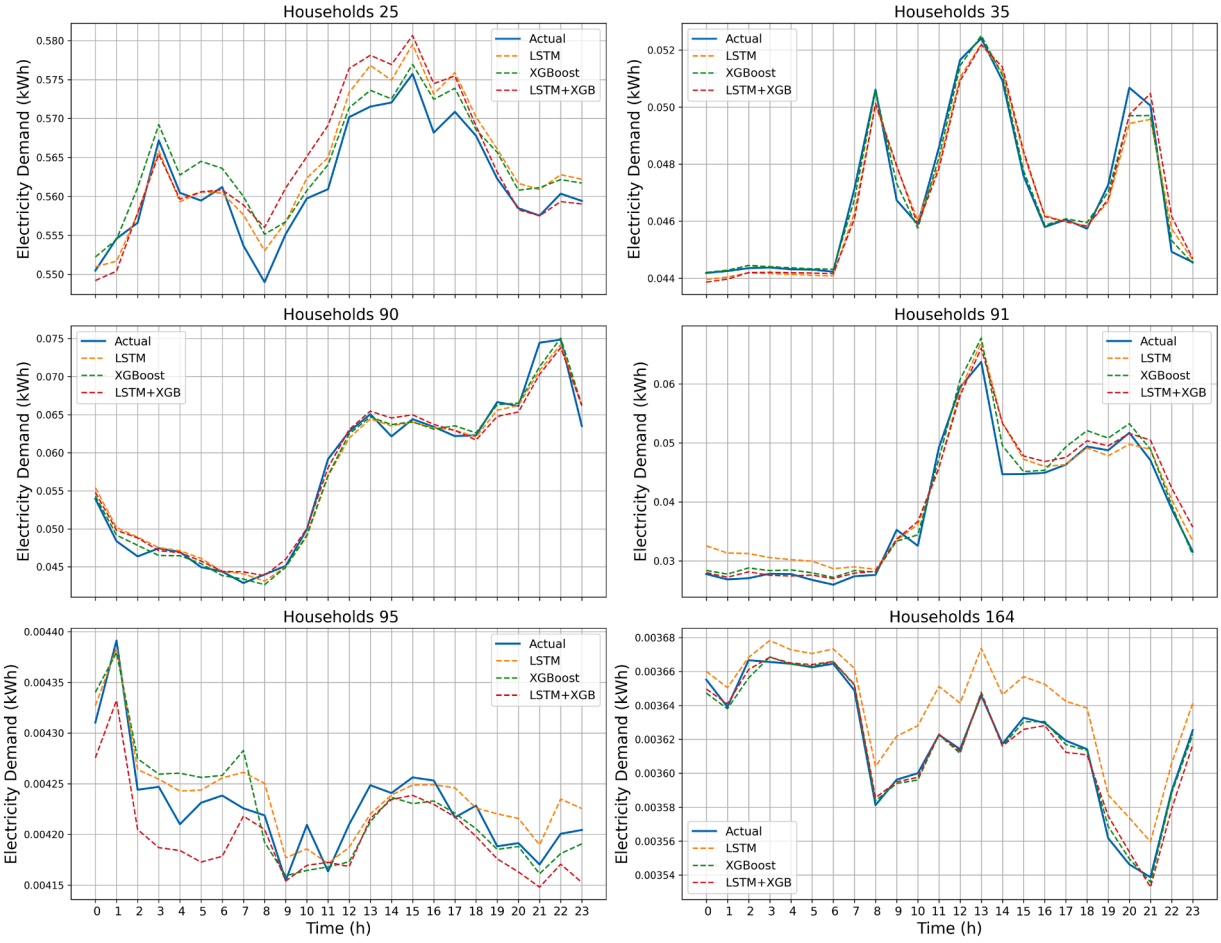


Fig. 12. Comparison of actual and predicted daily electricity demand for six representative households using XGBoost, LSTM, and Hybrid LSTM-XGBoost models.

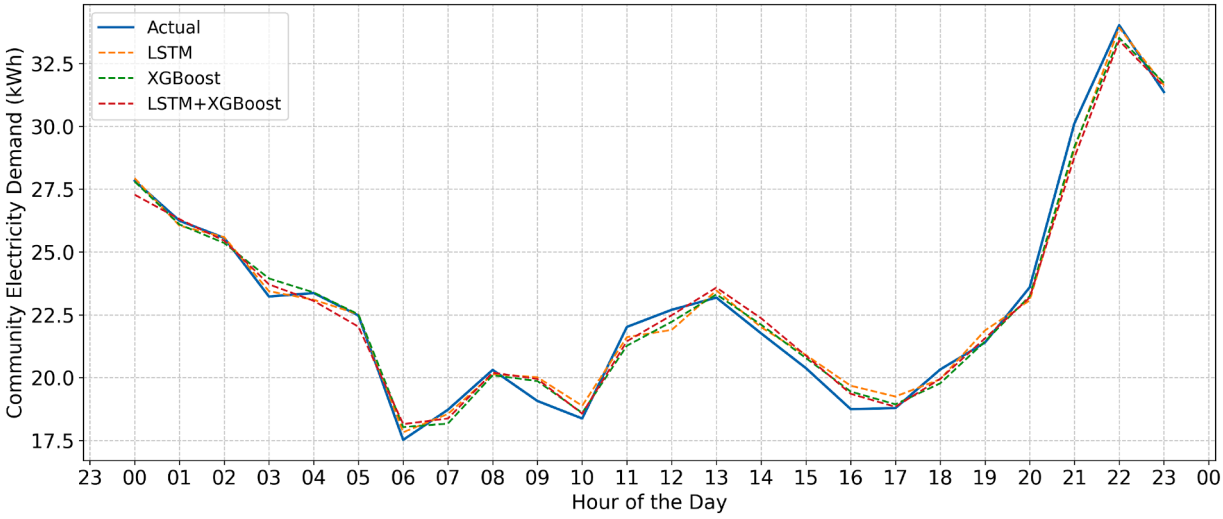


Fig. 13. Comparison of actual and predicted community demand on 25 May 2023.

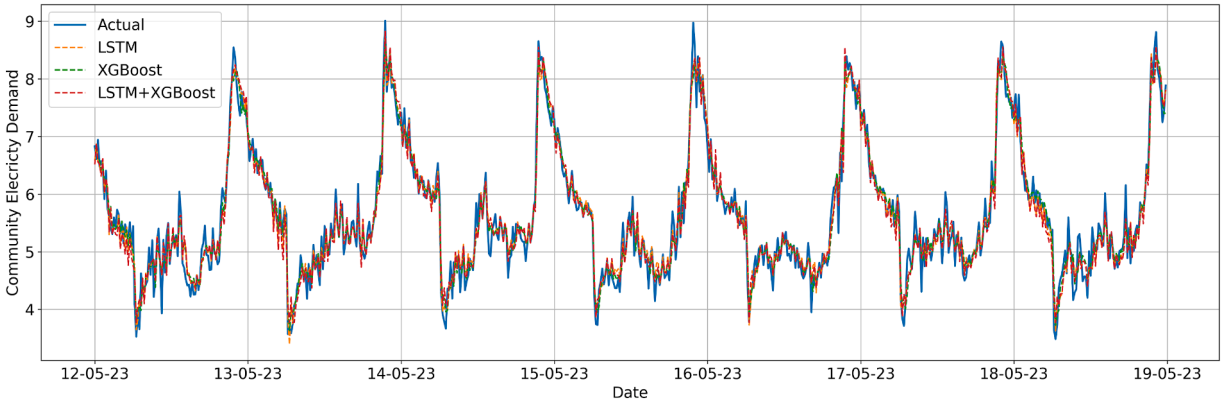


Fig. 14. Comparison of actual and predicted community demand for the week starting 12 May 2023.

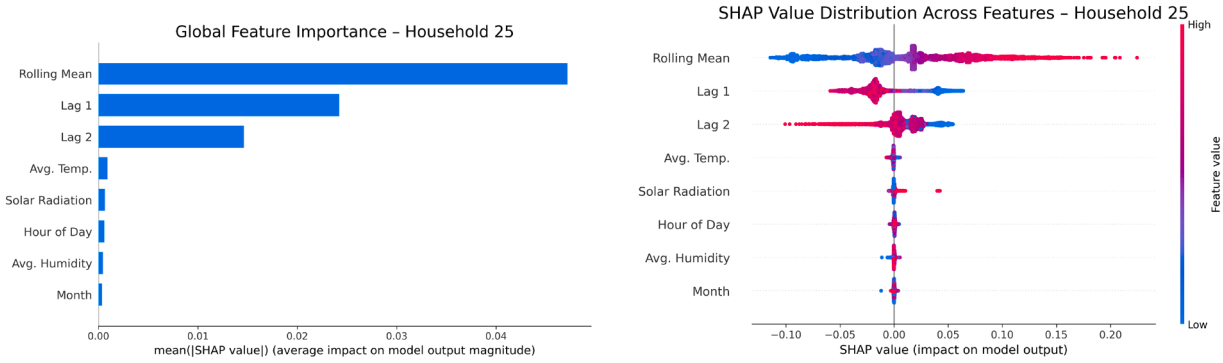


Fig. 15. SHAP analysis for Household 25: (a) global feature importance, (b) SHAP value distribution across features.

profiles highlight behavioural and contextual variability across users. For Household 25, short-term memory features, particularly rolling averages and lagged demand, dominate the model's decisions Fig. 15, indicating that recent consumption patterns are the primary drivers of forecast accuracy. Meteorological variables such as temperature and solar radiation play a comparatively minor role, suggesting limited weather sensitivity. In contrast, Household 164 Fig. 16 shows stronger dependence on temperature and humidity, alongside lag-based features, implying greater responsiveness to environmental conditions likely driven by climate-dependent appliance usage. At the community level Fig. 17, aggregation smooths out individual behavioral fluctuations, resulting in a more stable and predictable attribution pattern. Rolling averages and lag-24 demand remain the dominant predictors, while temporal encodings, such as hour of day, retain moderate importance. Meteorological variables contribute less prominently, as weather-induced variability averages out across all households. This illustrates how aggregation enhances temporal regularity and attenuates individual behavioural effects. Overall, the SHAP analysis reinforces the multi-level nature of electricity demand forecasting in RECs. Household-level forecasts reveal user-specific sensitivities to historical demand

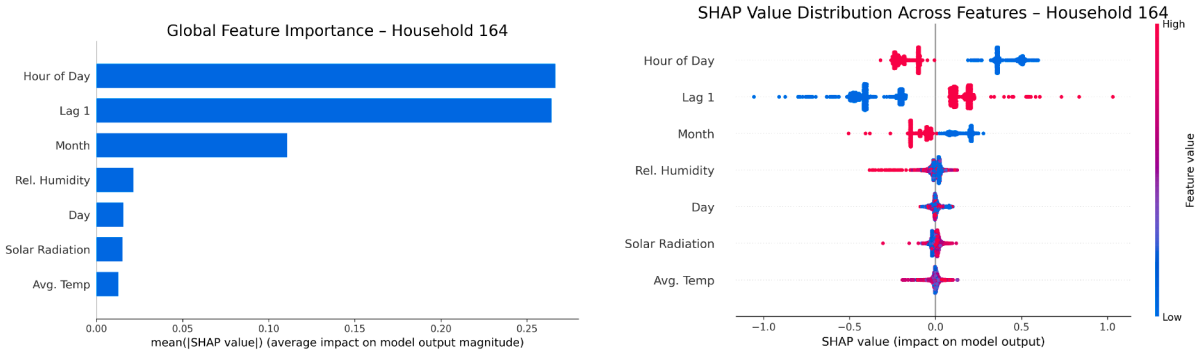


Fig. 16. SHAP analysis for Household 164: (a) global feature importance, (b) SHAP value distribution across features.

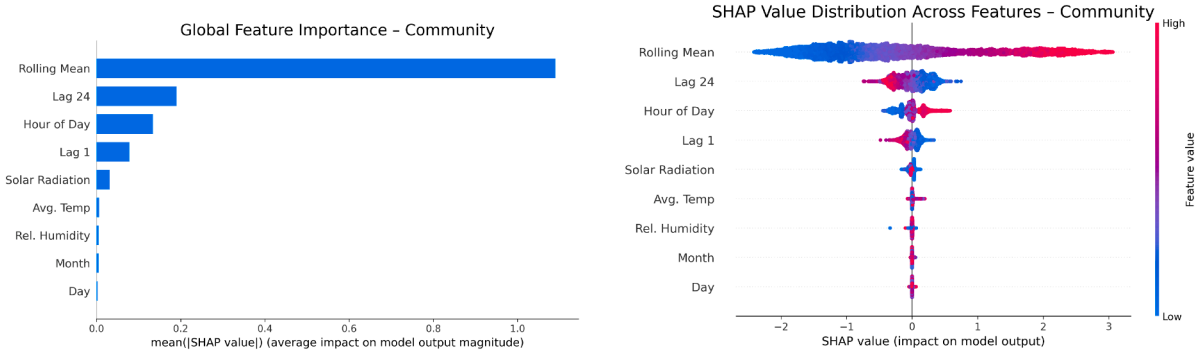


Fig. 17. SHAP analysis for community-level demand: (a) global feature importance, (b) SHAP value distribution across features.

and environmental drivers, community-level forecasts exhibit smoother, cyclical dependencies shaped by collective behavior. The LSTM successfully captures temporal dependencies before tree-based regression, as evidenced by the predominance of lagged demand characteristics among the most significant predictors. SHAP insights can support REC operations by identifying households with strong short-term memory dependence for battery scheduling or those sensitive to weather features for climate-adjusted tariffs. At the user level, dominant drivers revealed by SHAP can guide personalized feedback and engagement strategies. However, SHAP explanations are model-specific and may vary over time, reflecting associations rather than causal effects. These interpretable insights confirm the robustness of the hybrid model and provide actionable knowledge for demand-side management, battery scheduling, and local energy optimisation within RECs.

6. Discussion

The proposed hybrid LSTM–XGBoost framework demonstrates strong forecasting performance at both household and community levels within a REC. Using high-resolution 15-min data and integrating temporal, meteorological, and demand-derived features, the framework effectively captures cyclic consumption patterns as well as weather-driven variability. Across the conducted experiments, the hybrid configuration showed its clearest advantages at the community level and in selected household cases, achieving household-level MAPE as low as 1.25% and community-level MAPE of 3.14%, with corresponding R^2 values reaching 0.9558.These results compare favourably with existing studies on residential and REC load forecasting. Prior work has reported household-level MAPEs typically ranging between 5% and 10% when modelling stochastic and user-dependent demand behaviour using deep or hybrid architectures [35,36]. At the community scale, earlier REC-oriented studies [13,19] generally report MAPEs between 10% and 16%, often based on hourly resolution or synthetic datasets. By contrast, the present study leverages real-world 15-min measurements enriched with exogenous variables, enabling competitive household-level forecasting and strong community-level accuracy under REC operating conditions. These findings highlight the benefits of high temporal resolution and hybrid modelling for operational REC forecasting, particularly at the community level and under more variable demand conditions, rather than establishing universal superiority over alternative architectures.

Interpretability is a critical requirement for trustworthy REC operation. SHAP-based analysis indicates that short-term memory features, including lagged demand and rolling averages, dominate predictive behaviour across scales, while weather sensitivity varies across households. For example, Household 25 exhibits largely inertia-driven consumption patterns, whereas Household 164 shows stronger dependence on temperature and solar radiation. At the community level, aggregation attenuates individual weather effects, with demand dynamics primarily governed by lag-24 demand and temporal encodings. This multi-level interpretability enhances transparency by linking model outputs to physically and behaviourally meaningful drivers of consumption. From an operational perspective, household-level explanations can support targeted feedback and personalised demand-response strategies, while community-level insights inform storage scheduling, flexibility management, and coordination in local energy markets. Despite these encouraging results, several limitations should be acknowledged. Households with highly irregular or weakly correlated demand remain challenging to predict, indicating the potential value of adaptive retraining strategies or anomaly detection mechanisms. In addition, the current framework focuses on deterministic point forecasts; extending the approach to probabilistic forecasting would enable more risk-aware decision-making in REC operation. Finally, evaluating transferability across multiple communities represents an important direction for future research, particularly to reduce training overhead and improve generalisation in new deployments.

Overall, the proposed hybrid LSTM–XGBoost framework provides a practical and interpretable solution for short-horizon electricity demand forecasting in RECs. By combining high temporal resolution data, structured feature integration, and transparent hybrid modelling, it supports data-driven REC operation while remaining relevant for deployment un-

der context-dependent computational and retraining requirements. In addition to point forecasting accuracy, conformal prediction was employed to quantify forecast uncertainty, yielding well-calibrated prediction intervals with empirical coverage exceeding the nominal confidence level.

7. Conclusion

This study proposes a multi-scale forecasting framework that integrates LSTM-based temporal modelling with XGBoost within a deployment-oriented setting for real-world 15-min smart meter data from a Portuguese renewable energy community. The framework achieved strong predictive performance, with household-level mean absolute percentage errors as low as 1.25% in the best-performing cases. At the community scale, the model attained a coefficient of determination of 0.9558 with a mean absolute percentage error of 3.14%, indicating strong forecasting capability under REC operating conditions. These results support the effectiveness of the proposed framework in capturing both individual and collective consumption dynamics, while also highlighting the value of combining high-resolution data, multi-level forecasting, and interpretable modelling.

Beyond predictive performance, the framework enhances interpretability through Shapley additive explanations, which attribute the contribution of each feature to the model's predictions. The household-level analysis revealed diverse consumption drivers, ranging from demand inertia to weather sensitivity, while temporal and aggregated load features were dominant at the community level. Such interpretability supports transparency and stakeholder confidence, positioning the framework as an explainable decision-support tool for REC forecasting. From an operational perspective, the proposed forecasting framework provides actionable value for renewable energy community management. Household-level forecasts can enable targeted demand response participation and personalised energy feedback, while community-level predictions support peer-to-peer trading, storage scheduling, and local balancing.

Despite these encouraging findings, limitations remain. The empirical evaluation is based on a single renewable energy community, which may limit the generalisability of the framework to other contexts. Specific households with highly irregular and weakly correlated demand profiles posed forecasting challenges, indicating the need for household-specific model adaptation or anomaly detection. Moreover, the current implementation is centred on deterministic point forecasts, while forecast uncertainty was additionally examined through conformal prediction; broader probabilistic forecasting extensions remain an important direction for future work. Future research will therefore focus on adaptive retraining strategies for atypical households, broader cross-community validation, deeper integration with flexibility markets and energy trading platforms, and further development of uncertainty-aware forecasting methods. Collectively, these directions can strengthen the deployment readiness, interpretability, and operational value of forecasting frameworks in renewable energy communities.

Although this study focuses solely on forecasting, the generated short-term predictions are designed for integration into intelligent energy management systems in renewable energy communities. These forecasts can support downstream applications such as model predictive control, battery scheduling, load shifting, and peer-to-peer energy trading.

CRediT authorship contribution statement

Muhammad Akram: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Chiara Martone:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Giovanna Pallotta:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Maurizio Sasso:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Riccardo Amatore:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A. Bin-wise Sanity Check

Table A.6
Bin-wise sanity check comparing the average daily electricity demand of the randomly selected households with the distribution of all households within the same contribution class. Q1, Median, and Q3 denote the interquartile range of daily demand within each bin.

Contribution Bin	Selected HH	Q1 (kWh/day)	Median (kWh/day)	Q3 (kWh/day)	Selected Mean (kWh/day)	Within IQR
< 0.05%	164	0.0000	0.0000	0.1290	0.3326	No
0.05–0.25%	95	0.0000	0.7835	2.2260	1.5781	Yes
0.25–0.5%	35	1.2262	3.8285	5.6780	4.2682	Yes
0.5–1%	90	4.2450	7.5010	10.8860	6.8139	Yes
1–3%	91	8.2872	13.9700	22.7070	13.0504	Yes
> 3%	25	28.5112	51.4655	63.8265	43.2915	Yes

Nomenclature

Symbol	Description
y_t	Actual electricity demand at time t
$\hat{y}_t$	Predicted electricity demand at time t
$\mathbf{x}_t$	Input feature vector at time t (demand history, weather, temporal features)
f_k	kth regression tree in the XGBoost ensemble
$\mathcal{F}$	Functional space of regression trees
$J^{(k)}$	Objective function at iteration k in XGBoost
$\ell(\cdot)$	Loss function for a single sample (squared error)
$\Omega(f_k)$	Regularization term for tree complexity
T	Number of leaves in a regression tree
w_j	Weight of the jth leaf in a regression tree
γ, λ	Regularization coefficients (sparsity and weight penalty)
g_t	First derivative of the loss (gradient) w.r.t. prediction
h_t	Second derivative of the loss (Hessian) w.r.t. prediction
I_j	Set of training samples assigned to the jth leaf
G, H	Accumulated gradients (G) and Hessians (H) of a node
G_L, H_L	Accumulated gradients and Hessians of the left child node
G_R, H_R	Accumulated gradients and Hessians of the right child node
$\mathbf{f}_t$	Forget gate activation in LSTM at time t
$\mathbf{i}_t$	Input gate activation in LSTM at time t
$\mathbf{o}_t$	Output gate activation in LSTM at time t
$\mathbf{c}_t$	Cell state in LSTM at time t
$\tilde{\mathbf{c}}_t$	Candidate cell state in LSTM at time t
$\mathbf{h}_t$	Hidden state in LSTM at time t
$\mathbf{W}, \mathbf{U}, \mathbf{b}$	Weight matrices and bias vectors in LSTM gates
$\mathbf{W}_d, \mathbf{b}_d$	Dense projection layer weights and bias
$\mathbf{w}_y, \mathbf{b}_y$	Trainable regression parameters in LSTM output layer
$\mathbf{z}_t$	Latent feature vector produced by LSTM
$\hat{y}_t^{(\text{LSTM})}$	Prediction from standalone LSTM model at time t
$\mathcal{L}_{\text{LSTM}}$	Mean squared error (MSE) loss for LSTM training
$\sigma(\cdot)$	Sigmoid activation function
$\tanh(\cdot)$	Hyperbolic tangent activation function
$\odot$	Elementwise product operator
$\phi_\theta(\cdot)$	LSTM encoder mapping from input sequence to latent space
$F(\cdot)$	XGBoost regressor operating on embeddings and exogenous features
K	Number of trees in XGBoost ensemble
μ, σ	Mean and standard deviation used for feature standardization
N	Number of training samples
d	Dimensionality of the latent feature vector $\mathbf{z}_t$

References

[1] E. Gløersen, H. Gorny, M.M. Furtado, A. Münch, M. Alessandrini, C. Bettini, E.C. ofthe Regions, S. Foresight, ÖIR., t33, Implementing the European Green Deal: Handbook for Local and Regional Governments, CoR, 2022. https://books.google.pt/books?id=vYl6zwEACAAJ.

[2] United Nations Environment Programme, Executive summary 2022 global status report for buildings and construction: towards a zero-emission, efficient and resilient buildings and construction sector, 2022, https://wedocs.unep.org/20.500.11822/41134.

[3] European Union, Directive (EU) 2018/2001 of the European parliament and of the council of 11 December 2018 on the promotion of the use of energy from renewable sources (recast), 2018, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32018L2001. Brussels, Belgium.

[4] European Commission, Energy Communities, Energy communities, 2023, https://energy.ec.europa.eu/topics/markets-and-consumers/energy-communities_en.

[5] P. Esposito, E. Marrasso, C. Martone, G. Pallotta, C. Roselli, M. Sasso, M. Tufo, A roadmap for the implementation of a renewable energy community, Heliyon 10 (3) (2024) e28269. https://doi.org/10.1016/j.heliyon.2024.e28269

[6] F. Ceglia, P. Esposito, A. Faraudello, E. Marrasso, P. Rossi, M. Sasso, An energy, environmental, management and economic analysis of energy efficient system towards renewable energy community: The case study of multi-purpose energy community, J. Clean. Prod. 369 (2022) 133269. https://doi.org/10.1016/j.jclepro.2022.133269

[7] I.L.R. Gomes, M.G. Ruano, A.E. Ruano, From home energy management systems to communities energy managers: The use of an intelligent aggregator in a community in Algarve, Portugal, Energy Build. 298 (2023) 113588. https://doi.org/10.1016/j.enbuild.2023.113588

[8] Y. Wang, Y. Li, Y. Cao, W. Li, X. Yang, Y. Xu, H. Chen, M. Shahidehpour, Technical and practical application of integrated energy technology in smart community, Electr. Power Syst. Res. 232 (2024) 110422. https://doi.org/10.1016/j.epsr.2024.110422

[9] M. Ehtsham, G. Parisi, F. Pedone, F. Rossi, M. Zincani, E. Congiu, C. Marchionni, AI-powered advanced technologies for a sustainable built environment: a systematic review on emerging challenges, Sustainability 17 (17) (2025). https://doi.org/10.3390/su17178005

[10] Y. Chen, M. Guo, Z. Chen, Z. Chen, Y. Ji, Physical energy and data-driven models in building energy prediction: A review, Energy Rep. 8 (2022) 2656–2671. https://doi.org/10.1016/j.egyr.2022.01.162

[11] R. Ahmed, V. Sreeram, Y. Mishra, M.D. Arif, A review and evaluation of the state-of-the-art in PV solar power forecasting: techniques and optimization, Renew. Sustainable Energy Rev. 124 (2020) 109792. https://doi.org/10.1016/j.rser.2020.109792

[12] Y. Lin, J. Liu, K. Gabriel, W. Yang, C.-Q. Li, Data-driven based prediction of the energy consumption of residential buildings in Oshawa, Buildings 12 (11) (2022). https://doi.org/10.3390/buildings12112039

[13] M. Matos, J. Almeida, P. Gonçalves, F. Baldo, F.J. Braz, P.C. Bartolomeu, A machine learning-based electricity consumption forecast and management system for renewable energy communities, Energies 17 (3) (2024) 630. https://doi.org/10.3390/en17030630

[14] L. Baur, K. Ditschuneit, M. Schambach, C. Kaymakci, T. Wollmann, A. Sauer, Explainability and interpretability in electric load forecasting using machine learning techniques - a review, Energy and AI 16 (2024) 100358. https://doi.org/10.1016/j.egyai.2024.100358

[15] Y. Özüpak, S. Mansurov, Optimizing electricity demand forecasting with a novel RNN-LSTM hybrid model, Energy Sources, Part B: Econ. Plann. Policy 20 (1) (2025) 2531448. https://doi.org/10.1080/15567249.2025.2531448

[16] Y. Özüpak, S. Mansurov, A hybrid deep learning approach for multi-output short-term electricity demand forecasting, Concurrency and Comput. Pract. Exper. 37 (25-26) (2025) e70356. https://doi.org/10.1002/cpe.70356

[17] J.W. Chan, C.K. Yeo, A Transformer based approach to electricity load forecasting, Electr. J. 37 (2024) 107370. https://doi.org/10.1016/j.elecj.2024.107370

[18] C. Antonesi, et al., Hybrid transformer model with liquid neural networks and learnable encodings for buildings' energy forecasting, Energy AI 20 (2025) 100489. https://doi.org/10.1016/j.egyai.2025.100489

[19] L. Semmelmann, S. Henni, C. Weinhardt, Load forecasting for energy communities: a novel LSTM-XGBoost hybrid model based on smart meter data, Energy Inf. 5 (Suppl 1) (2022) 24. https://doi.org/10.1186/s42162-022-00212-9

[20] H. Kim, S. Park, S. Kim, Time-series clustering and forecasting household electricity demand using smartmeter data, Energy Rep. 9 (2023) 4111–4121. https://doi.org/10.1016/j.egyr.2023.03.042

[21] K. Dab, N. Henao, S. Nagarsheth, Y. Dubé, S. Sansregret, K. Agbossou, Consensus-based time-series clustering approach to short-term load forecasting for residential electricity demand, Energy Build. 299 (2023) 113550. https://doi.org/10.1016/j.enbuild.2023.113550

[22] E.T. Habtemariam, M.M. Ballesteros, A. Troncoso, F.M. Álvarez, A novel approach based on clustering and optimized ensemble deep learning for energy consumption forecasting in Ethiopia, Neurocomputing 637 (2025) 130027. https://doi.org/10.1016/j.neucom.2025.130027

[23] R. Akter, M.G. Shirkoohi, J. Wang, W. Mérida, An efficient hybrid deep neural network model for multi-horizon forecasting of power loads in academic buildings, Energy Build. 329 (2025) 115217. https://doi.org/10.1016/j.enbuild.2024.115217

[24] L. Botman, J. Lago, T. Becker, K. Vanthournout, B. De Moor, A global probabilistic approach for short-term forecasting of individual households electricity consumption, Appl. Energy 382 (2025) 125168. https://doi.org/10.1016/j.apenergy.2024.125168

[25] X. Zhao, P. Duan, X. Cao, Q. Xue, B. Zhao, J. Hu, C. Zhang, X. Yuan, A probabilistic load forecasting method for multi-energy loads based on inflection point optimization and integrated feature screening, Energy 327 (2025) 136391. https://doi.org/10.1016/j.energy.2025.136391

[26] R. Wang, H. Yun, R. Rayhana, J. Bin, C. Zhang, O.E. Herrera, Z. Liu, W. Mérida, An adaptive federated learning system for community building energy load forecasting and anomaly prediction, Energy Build. 295 (2023) 113215. https://doi.org/10.1016/j.enbuild.2023.113215

[27] J. Coignard, M. Janvier, V. Debusschere, G. Moreau, S. Chollet, R. Caire, Evaluating forecasting methods in the context of local energy communities, Int. J. Electr. Power Energy Syst. 131 (2021) 106956. https://doi.org/10.1016/j.ijepes.2021.106956

[28] X. Li, L. Ma, P. Chen, H. Xu, Q. Xing, J. Yan, S. Lu, H. Fan, L. Yang, Y. Cheng, Probabilistic solar irradiance forecasting based on XGBoost, Energy Rep. 8 (2022) 1087–1095. ICPE 2021 - The 2nd International Conference on Power Engineering. https://doi.org/10.1016/j.egyr.2022.02.251

[29] A. Miraki, P. Parviainen, R. Arghandeh, Probabilistic forecasting of renewable energy and electricity demand using graph-based denoising diffusion probabilistic model, Energy AI 19 (2025) 100459. https://doi.org/10.1016/j.egyai.2024.100459

[30] N. Mendes, J. Mendes, J. Mohammadi, P. Moura, Federated learning framework for prediction of net energy demand in transactive energy communities, Sustainable Energy Grids Netw. 40 (2024) 101522. https://doi.org/10.1016/j.segan.2024.101522

[31] E. Sarantinopoulos, V. Michalakopoulos, E. Sarmas, V. Marinakis, L. Toderean, T. Cioara, Meta-heuristic federated learning aggregation methods for load forecasting, Energy AI 22 (2025) 100594. https://doi.org/10.1016/j.egyai.2025.100594

[32] F. Monteiro, R. Oliveira, J. Almeida, P. Gonçalves, P. Bartolomeu, J. Neto, R. Deus, Electricity consumption dataset of a local energy cooperative, Data Brief 54 (2024) 110373. https://doi.org/10.1016/j.dib.2024.110373

[33] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

[34] T. Chen, C. Guestrin, XGBoost: a scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794. https://doi.org/10.1145/2939672.2939785

[35] T.-Y. Kim, S.-B. Cho, Predicting residential energy consumption using CNN–LSTM neural networks, Energy 182 (2019) 72–81. https://doi.org/10.1016/j.energy.2019.05.230

[36] W. Cao, H. Liu, X. Zhang, Y. Zeng, Residential load forecasting based on long short-term memory considering temporal local attention, Sustainability 16 (24) (2024) 11252. https://doi.org/10.3390/su162411252